



**Verified Carbon
Standard**

ELMADAG BIOGAS POWER PLANT

Document Prepared by



Climate Solutions
REFRAME THE FUTURE

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Elmadag Biogas Power Plant (hereafter referred to as the Project) is a biogas-to-energy project that captures biogas from animal manure, through anaerobic digestion and utilizes it to generate electricity via gas engines. The animal manure is collected from the neighbouring animal farms with special trucks and transferred to the biogas plant. Continuously Stirred Tank Reactor (CSTR) type anaerobic digesters are used to capture methane-rich biogas. Biogas then utilized in gas engines for heat and electricity generation. Waste heat from gas engines is mainly utilized for heating anaerobic fermenters. The electricity generated two biogas engines installed at the plant with the total capacity of 2.406 MWe is not only supplied to the national grid, but also used for auxiliary/internal consumption required for the plant to function.

The project is being implemented by ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ, the Project Owner, in Elmadağ district, Ankara province, in Turkey. The start date of the project is determined as December 31, 2020.

Prior to the project activity, livestock farms supplying animal manure to the biogas plant were storing the manure generated at their farms in uncovered anaerobic lagoons for a retention period of 2-3 months and then, generally applying it to land, as fertilizer, directly without any proper treatment. Untreated animal manure has, however, negative impacts on both the environment and human health^{1,2,3,4}. In addition, untreated animal manure is an important source of methane which is one of the most potent Greenhouse Gas (GHGs).

This project adopts the renewable crediting period of 7 years, twice renewable for a total of 21 years. The estimated average net annual emission reductions are 32,857 tCO₂eq/y. Accordingly, the project is expected to generate 229,999 tCO₂eq emissions reduction during the first crediting period.

¹ https://www.researchgate.net/profile/Ramasamy-Rajesh-Kumar/publication/257840261_Application_and_Environmental_Risks_of_Livestock_Manure/links/5619224a08ae78721f9cfd93/Application-and-Environmental-Risks-of-Livestock-Manure.pdf (Accessed on 20.10.2022)

² https://apps.who.int/iris/bitstream/handle/10665/75700/1/9789241564519_eng.pdf (Accessed on 20.10.2022)

³ <https://pubs.acs.org/doi/abs/10.1021/acs.est.7b02811> (Accessed on 20.10.2022)

⁴ <https://www.sciencedirect.com/science/article/pii/S0048969717333120> (Accessed on 20.10.2022)

Audit Type	Period	Program	WVB Name	Number of years
Validation	26-December-2022	Verified Carbon Standard (VCS)	EcoLance Pvt. Ltd.	1

1.2 Sectoral Scope and Project Type

The proposed project activity is a biogas project, capturing Greenhouse Gases (GHGs), methane in particular, generated from animal manure that would be otherwise left to decay in anaerobic conditions at the neighboring farms by utilizing/treating animal manure in CSTR-type anaerobic digesters at the project site, mainly for electricity generation. Thus, the project falls into sectoral scope 01: Energy industries (renewable -/ non-renewable sources) (Type I Component), and sectoral scope 13: Waste handling and disposal (Type III Component) ⁵.

The project is not a grouped project.

1.3 Project Eligibility

The project is a biogas project that involves capturing GHGs, methane in particular, from animal manure in anaerobic conditions and utilization of this biogas via gas engines, for electricity generation. As per the scope defined under Section 2.1.1 of the VCS Standard 4.5⁶, the eligibility of the project is assessed as follows:

1) The seven Kyoto Protocol greenhouse gases:

In the project activity, manure generated at the neighbouring animal farms are collected and treated to capture the methane (CH₄) and nitrous oxide (N₂O), which are among the seven Kyoto Protocol GHGs. The project employs four anaerobic digestion tanks and two gas engines for methane capture and electricity generation and avoids the methane emissions from uncovered anaerobic lagoons present in the baseline scenario. Thus, the project is eligible under this scope.

2) Ozone-depleting substances:

Not applicable to the project.

⁵ <https://verra.org/project/vcs-program/projects-and-jnr-programs/vcs-sectoral-scopes/>

⁶ <https://verra.org/wp-content/uploads/2023/08/VCS-Standard-v4.5.pdf> (Release Date 29/08/2023) (Accessed on 08/12/2023)

- 3) Project activities supported by a methodology approved under the VCS Program through the methodology approval process:

Not applicable to the project.

- 4) Project activities supported by a methodology approved under an approved GHG program, unless explicitly excluded (see the Verra website for exclusions).

The project applies two small-scale approved CDM baseline and monitoring methodologies. The related methodologies are as follows:

- AMS-III.D.: “Methane recovery in animal manure management systems”, Version 21.0⁷
- AMS-I.D.: “Grid connected renewable electricity generation”, Version 18.0⁸

- 5) Jurisdictional REDD+ programs and nested REDD+ projects as set out in the VCS Program document Jurisdictional and Nested REDD+ (JNR) Requirements.

Not applicable to the project.

The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal or destruction. The VCS Program also excludes the project activities under the circumstances indicated in Table 1, of the VCS Standard (v4.5). Based on this table, this project activity is not among the excluded ones.

1.4 Project Design

This project is designed to include a single installation only.

Eligibility Criteria

The project is not a grouped project.

1.5 Project Proponent

Organization name	CS Climate Solutions Danışmanlık A.Ş. (herein after Climate Solutions)
Contact person	Dr. Umut ÖNDER
Title	Chairperson (Project Proponent/Consultant/Account Manager)

⁷ <https://cdm.unfccc.int/methodologies/DB/H9DVS2407GEZQYLYNWUX23YS6G4RCv> (Accessed on 04/07/2023)

⁸ <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK> (Accessed on 04/07/2023)

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Organization name	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ
Contact person	Mr. Başar Beyazoğlu
Title	Project Owner
Address	Ilkbahar Mahallesi, 616 Sokak No:8 Çankaya/ANKARA - TÜRKİYE
Telephone	+90 216 403 10 40
Email	info@renesco.com.tr

1.6 Other Entities Involved in the Project

N/A.

1.7 Ownership

Both the approval of the Environmental Impact Assessment (EIA) issued by the Ministry of Environment, Urbanization and Climate Change in Turkey (given in Appendix-4) and the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey (given in Appendix-2) establishes the **project ownership** to **ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ**. These official documents accordingly establish the property and contractual rights of both the plant and equipment that generate GHG emission reductions. Nevertheless, it is important to note that Dr. Umut ÖNDER, the Chairperson of Climate Solutions, is acting as a designated Carbon Manager of ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ as per the agreement between these two parties, a document that is shared with the VVB in time of validation.

1.8 Project Start Date

The project start date, as defined on the VCS Program Definitions 4.4⁹, is the date on which the project began generating GHG emission reductions or removals.

The chronology of the project as given in Table 1, starts with the production license issued by Energy Market Regulatory Authority (EMRA) in 01.08.2019, which is necessary for renewable energy-based electricity generation plants to enable their activities such as electricity generation, transmission, distribution, wholesale/retail sale in the market.¹⁰ Following that, the positive decision Environmental Impact Assessment (EIA) was obtained on 01.03.2017. The installation of gas engines was completed on 26.10.2020. The provisional acceptance to generate electricity was obtained by the Ministry of Energy and Natural Resources in 31.12.2020. Therefore, this date is considered as the project start date, since, in alliance with the definition of VCS, by this date the project technically and officially began to generate GHG emission reductions.

Table 1. Project Chronology

Date	Project Milestones
01/03/2017	EIA Positive Decision
01/08/2019	EMRA Production License
26/10/2020	Installation of Gas Engine
31/12/2020	Provisional Acceptance to Generate Electricity (Start Date of the Project)

1.9 Project Crediting Period

The VCS Standard 4. 5¹¹ states that the project crediting period for non-AFOLU projects shall be either seven years, twice renewable for a total of 21 years, or ten years fixed. Since this project is a non-AFOLU project, the renewable crediting period of 7 years, twice renewable for a total of 21 years is adopted by the Project Proponents (PPs). The first crediting period starts on 31.12.2020 and ends on 30.12.2027 (the start and end dates included).

⁹ VCS Program Definitions v4.4 (Release date: 29 August 2023) <https://verra.org/wp-content/uploads/2023/08/VCS-Program-Definitions-v4.4.pdf> (Accessed on 08/12/2023)

¹⁰ <https://enerji.gov.tr/eigm-yenilebilir-enerji-uretim-faaliyetleri-lisansli-elektrik-uretimi>

¹¹ <https://verra.org/wp-content/uploads/2023/08/VCS-Standard-v4.5.pdf> (Released Date: 29/08/2023)) (Accessed on 08/12/2023)

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO _{2e})
31/12/2020	90
01/01/2021-31/12/2021	32,857
01/01/2022-31/12/2022	32,857
01/01/2023-31/12/2023	32,857
01/01/2024-31/12/2024	32,857
01/01/2025-31/12/2025	32,857
01/01/2026-31/12/2026	32,857
01/01/2027-30/12/2027	32,767
Total estimated ERs	229,999
Total number of crediting years	7
Average annual ERs	32,857

1.11 Description of the Project Activity

The project received the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey on August 01, 2020, for the maximum operational licensed capacity of 2.406 MWe (Please see Appendix-2). The project has obtained the

approval of the Environmental Impact Assessment (EIA) issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on March 01, 2017 (Please see Appendix-4).

The project has become operational by starting to generate electricity and supplying it partly to the Turkish national grid on December 31, 2020 (Please see Appendix-3), which is the project start date.

This project involves biogas production through the biodegradation of animal manure in an anaerobic environment. The biogas that is released during the biodegradation of animal manure is used for electricity and heat production via cogeneration units. The proposed project activity currently has two installed biogas engines at the biogas power plant, with each capacity of 1,203 KWe. The waste heat by the gas engines is mainly used to maintain the required digester temperature. On the other hand, most of the generated electricity is directly fed to the national grid, whilst some is used for auxiliary electricity consumption required at the project site.

Prior to the project activity, animal manure generated at the supplier farms were left to decay anaerobically in uncovered anaerobic lagoons for an average retention period of 2-3 months. This animal manure management system led to the release of methane directly into the atmosphere.

The project generates GHG emission reductions through the capture of the methane gas generated in the animal farms as well as the replacement/avoidance of fossil-fuel based electricity connected the national grid by supplying biogas-based electricity.

Because the animal manure is left to decay in uncovered anaerobic lagoons without any proper treatment in the baseline scenario, none of the equipment installed within the project boundary exists in the baseline scenario.

The equipment or units involved in the project are given in Table 2. Also, the flow chart of the Elmadag Biogas Power Plant is given in Figure 1 below.

The project consists mainly of the installation of administrative building; parking area; wheel washing unit; weighbridge; animal manure receiving tanks/units; fermenter tanks/anaerobic digesters; hygienization tank (and drying unit); separators; liquid fermented product storage unit; solid fermented product warehouse; gas engines; desulphurization system; biogas burning chimney (flare); engine room; electricity transmission-monitoring and control units; and plant and process monitoring-control and management systems.

Table 2. Units/Equipment/Facilities at the Project Site

UNIT NAME	NUMBER OF UNITS (Unit)
Administrative Building	1
Parking Area	1
Weighbridge	1
Wheel Washing unit	1
Animal Manure Receiving Tanks /Units	5
Fermenter Tanks / Anaerobic Digesters	4
Separators	2
Liquid Fermented Product Storage Unit	1
Solid Fermented Product Warehouse	1
Gas Engines	2
Desulphurization System	1
Biogas Burning Chimney (Flare)	1
Engine room	1

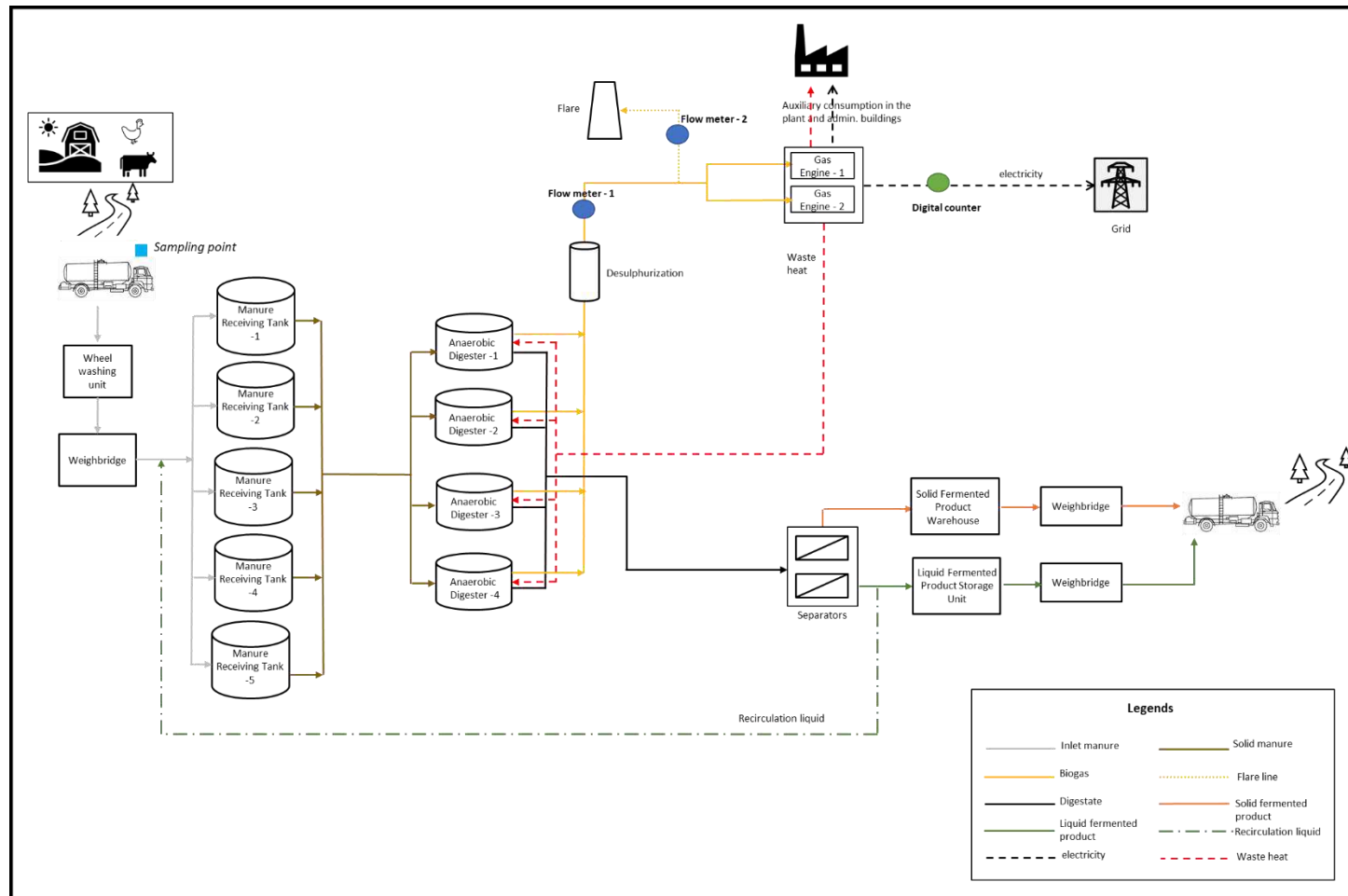


Figure 1. Flow Chart of Elmadag Biogas Power Plant

Emissions associated with the baseline scenario are the CH₄ emissions due to the atmospheric release of methane sourced from animal manure left to decay anaerobically and CO₂ emissions due to the power generation by the national grid which is dominated by fossil fuel-burning power plants. Through the project's implementation, the methane previously released from animal manure left to decay in uncovered anaerobic lagoons is captured via anaerobic digestion and utilized to produce thermal and electric energy through biogas generator systems in the biogas power plant. In addition, the electricity generated at the plant is partly fed to the national grid, avoiding and/replacing the dispatch of the same amount of electricity from fossil-fuel-based power plants connected to the national grid. Thus, none of the equipment installed within the project boundary exists in the baseline scenario.

Weighbridge:

There is one weighbridge (ESIT Brand) with a capacity of 60 tons at the biogas plant. The weighbridge is built at the entrance of the plant to continuously record the type and the quantity of animal manure entering the plant to be fed to ADs, as well as the name of the farms, as their sources. In addition, the type, the quantity, and the final destinations of fermented products leaving the plant to be applied to agricultural plots are also continuously recorded in the weighbridge records logbook by the staff in charge of the weighbridge.

The weighbridge installed at the plant is given in Figure 2.



Figure 2. Weighbridge

Wheel and Vehicle Washing Unit:

The wheel and vehicle washing system is built at the entrance of the plant to prevent the vehicles entering and leaving the facility from carrying diseases, particularly animal-borne ones, inside and outside of the facility.

Animal Manure Receiving Tanks/Units:

Five animal manure receiving tanks were installed for collection and preparation of animal manure before transferring to digesters. The receiving tank is made of C30 concrete with impermeable insulation. This unit is used for preliminary acceptance of animal manure to the biogas plant as well as for maintaining the required dry matter balance of the manure. Animal manure is accepted to the facility's waste receiving units on a daily basis through vacuum trucks. Necessary control systems are established to monitor the manure transfer.

Fermenter Tanks / Anaerobic Digesters:

Anaerobic fermentation is known to be an efficient animal manure management that generates biogas as product. Biogas is a clean energy source that is used in areas such as heat and electricity generation by direct and indirect usage methods. Biogas has become one of the most important renewable sources of energy in different parts of the world that draws more attention in both developed and developing countries day by day.

Within the scope of the project, there are four digesters installed where the animal manure transferred from the pre-pool is collected and the animal manure is fermented. The capacities of the digesters in the plant are 6,000 m³. It has been covered with EPDM membrane and water and heat insulation are performed. There is a heating system in the fermenter where the hot water obtained from the biogas generator system is circulated. The length of pipes required for the heating system are determined according to the heat loss calculation, and the fittings are installed on the base and side walls of the fermenter units. Heat insulation has been made on the ground and exterior of the fermenter to protect the heat inside. The technical details and operating conditions of the fermenter tanks/anaerobic digesters are given below:

Operating Conditions for the Fermenter Tanks/Anaerobic Digesters Reactor								
Unit name	Type of Digester	Unit Amount	Capacity (m ³)	Isolation thickness	Temp. (°C)	pH	C/N Ratio	Retention Time (days)
Anaerobic Digester	EPDM Membrane Enclosed Anaerobic Digester	4	6,000	100 mm	35 – 37	6.5 – 8	30/1	35 – 45

The fermenter tanks/anaerobic digesters in the Elmadag Biogas Power Plant are shown in Figure 3 below.



Figure 3. A fermenter tank/anaerobic digester of the project

Anaerobic digestion technology has been implemented in the facility. Anaerobic digestion occurs in the fermenter tanks/anaerobic digesters of the plant as part of the project activities. Biogas production is a combination of microbiological and biochemical processes. Microbiological processes of anaerobic fermentation consist of hydrolysis, acidogenesis, acetogenesis, and methanogenesis. Each of these stages is carried out by three different groups of microorganisms; fermentative bacteria, anaerobic oxidant bacteria, and methanogenic archaea, respectively.

In the first two stages (hydrolysis and acidogenesis); polymers (carbohydrates, proteins, and fats etc.) are converted into monomers (long-chain fatty acids, glycerol, amino acids, sugars, etc.), which will be converted into short-chain fatty acids, alcohols, hydrogen, and carbon dioxide through fermentation reactions, in the next step.

In the next step (acetogenesis), acetate, carbon dioxide and hydrogen are formed, which will be used by the methanogens in the final stage of anaerobic fermentation, through anaerobic oxidation of and acids alcohols.

At the end of the anaerobic fermentation process, methane is produced by methanogens.

The anaerobic fermentation process is illustrated in Figure 3 below.

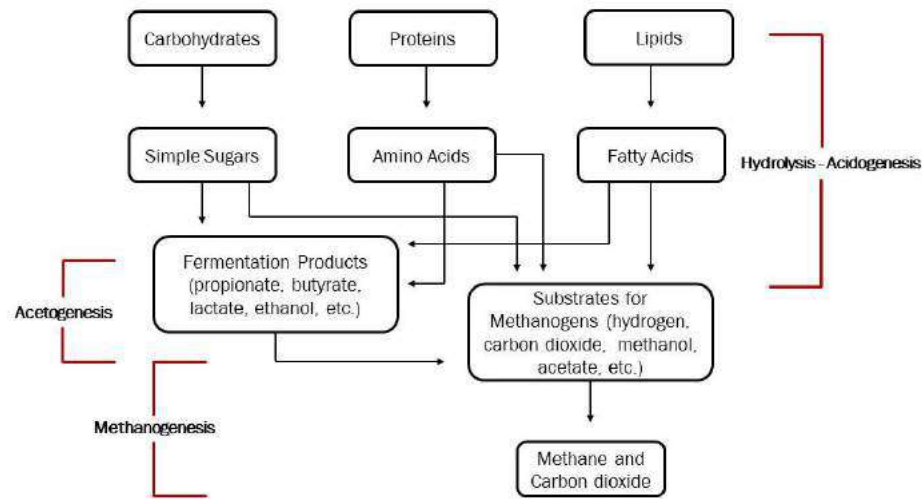


Figure 4. Flow Chart of the Anaerobic Digestion Process¹²

Separators:

There are two separators with 5.5-7.5 kW power within the scope of the project. The separators are used to separate the solid product from the liquid product. The sieve diameter of the separators is 700 microns and operates according to the squeezing filter principle to be transformed into final product in liquid and solid form.



Figure 5. Separators

¹² The flowchart has been modified from study of the Gavala et al., (2002). <https://link.springer.com/book/10.1007/3-540-45839-5?noAccess=true> (page 59) accessed on 13.06.2022

Liquid Fermented Product Storage Unit:

There is one liquid fermented product storage unit installed at the project site. The fermented product can be used as a liquid or a solid by dehydrating. Products with a dry matter content of up to 15% are considered liquid fermented products. In case the liquid fermented product is dehydrated, the oxygenated process is applied to ensure that the solid product obtained provides the stability values as per Annex-413 of the Communiqué¹⁴.

Solid Fermented Product Warehouse:

There is one solid fermented product warehouse installed at the project site. As per the Communiqué, solid fermented product coming out of the solid fermented product hygiene and drying unit, a storage area is built, in a size where these products will be deposited for minimum 1 months without being affected by meteorological events.

Gas Engines:

The biogas plant has two gas engines (JENBACHER Brand) each with a capacity of 1,203 kWe. The gas produced in the reactor is cooled in a humidifier and the excess moisture is removed. Here, it is transferred to the gas engine with the help of a blower. In the gas engine, there is a system that is called the gas train and evaluates the parameters related to the gas in its content. The purpose of this is not to damage the gas engine and to get 100% performance. Combustion occurs after the gas passing through here is mixed with air in the gas engine. The mechanical energy generated here is transferred to the alternator by the shaft to convert it into electrical energy. Mechanical energy is converted into electrical energy here and sent first to the transformers and then to the network line. The gas engines installed at the plant is given in Figure 6.

Table 3. Output / Fuel Consumption of the Gas Engine

Brand / Model	JENBACHER TA1100-0111	
Power according to ISO Standards ICFN	kWe	1,203
Effective fuel pressure and nominal speed	Bar	20.21
Fuel Type		Biogas
Methane content / Minimum methane content	MZ	135 (117 d)
Compression Ratio	Epsilon	12.5

¹³ <https://www.resmigazete.gov.tr/eskiler/2015/10/20151010-3-1.pdf> (Accessed on 08/12/2023)

¹⁴ <https://www.resmigazete.gov.tr/eskiler/2015/10/20151010-3.htm> (Accessed on 08/12/2023)

Min. / Max. gas pressure (gas train inlet)	mbar	90 -200 c)
Gas pressure max. fluctuation	mbar/sec	10
Section 2 inlet temperature	°C	55
Specific engine fuel consumption	g/kWh	0.20
Max. oil temperature	°C	85
Jacket water temperature max.	°C	95
Oil tank capacity (filling)	Lt	~437

Table 4. Technical Data of Gas Engines

Manner of work		4 cycles
Configuration		V 70
Number of cylinders		20
Diameter	mm	145
Piston travel	mm	185
Nominal speed	rpm	1,500
Average piston speed	m/s	9.25
Length	mm	3,750
Width	mm	1,580
Height	mm	2,033
Empty weight	kg	7,200
Gross weight	kg	7,900
Moment of inertia	kgm ²	11.64
Rotation direction (from flywheel view)		Left
Radio interference level VDE 0875		N
Starter motor output	kW	13
Starter motor voltage	V	24



Figure 6. Gas Engines

Desulphurization System:

There is one desulphurization system (consisting of two gas scrubbers) at the biogas plant. The biogas obtained is taken to the desulphurization device by blower. Desulphurization system shall reduce the unwanted H_2S in the biogas below 200 ppm and at the same time keep the biogas constant at the desired 150 mbar pressure at the zero-pressure regulator inlet of the biogas engine.

The desulphurization system installed at the plant is given in Figure 7.



Figure 7. Desulphurization system

Biogas Burning Chimney (Flare):

There is one ENT Makine brand open biogas burning chimney (flare) at the biogas plant with a thermal power of 1100 kW and height of 5.7 m. Flare is operated to stabilize the gas in the safest way in case of emergency or in the case that the gas cannot be combusted in the gas engines or stored.

Wastewater Management:

As per the discharge standards set out in the Water Pollution Control Regulation published in the official Gazette dated 31/12/2004 and numbered 25687¹⁴, the domestic wastewater is collected within the facility through municipal sewage trucks by municipal authorities and is transported to the nearest wastewater treatment plant.

¹⁴ <https://www.resmigazete.gov.tr/eskiler/2004/12/20041231.htm>

1.12 Project Location

The project is located in Elmadağ district, Ankara province, Turkey. The central coordinated of the project location are longitude of + 40.037864 and latitude of +33.114366.

The location of the project is illustrated in Figure 8 below. Also, Elmadağ Biogas Power Plant images are given in Figure 9 and Figure 10.



Figure 8. Location of the Project (red spotted)



Figure 9. Entrance of Elmadağ Biogas Power Plant



Figure 10. Elmadag Biogas Power Plant

Not only the names but also the coordinates of the farms are, accordingly, given in Table 5.

Table 5. Central Locations of the Farms Supplying Manure to the Project

FARM'S NAME ¹⁵	Livestock Types (L)	LATITUDE	LONGITUDE	DISTANCE (km)
Ali Bey Çiftliği (Necmettin Erden)	Other Cattle	40,03514	33,073257	7.3
Erol Yılmaz	Other Cattle	39,96814	33,20670	10.6
Bey Besi (Veysel Karabey)	Other Cattle	40,02210	33,05187	15.3
Şenol Yılmaz Besi Çiftliği	Other Cattle	39,95783	33,21770	12.2
Saygın Bey (Veysel Karabey)	Other Cattle	40,04001	33,11034	3.6
Aytekin Besi	Other	39,96402	33,22377	17.1

¹⁵ The provided farms represent the current manure suppliers of the project in time of preparation this PD. However, the project is expected to supply animal manure from farms might be included (hens and/or cattle farms) or some might be excluded from the project within the crediting period of the project activity. These potential changes might be dependent on the activities of the farms, manure quality, or the agreements between the farm owners and the project owner. The changes, if any, in this list will be reported as a project description deviation in the upcoming monitoring periods and provided to the VVB.

	Cattle			
Has Besi	Other Cattle	39,95732	33,21872	12.3
İzzet Alp	Other Cattle	39,97836	33,01407	26.8
yılmazlar besi (kerim yılmaz)	Other Cattle	39,90155	32,89994	23.1

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity is as follows:

Before the project activity, cattle manure generated at neighbouring animal farms which supply manure to the biogas plant were left to decay in anaerobic conditions within uncovered anaerobic lagoons for an average retention period of 2-3 months; then, directly applied to land without any proper treatment. Hence, greenhouse gases generated from animal manure were released directly into the atmosphere.

In addition, the quantity of electricity produced using biogas technologies would have been produced by power plants connected to the national grid in Turkey which is heavily dependent on fossil fuel resources¹⁶.

The baseline scenario is the same as the scenario existing prior to the start of the implementation of the project activity. Please refer to Section 3.4 of VCS PD for further details.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The two main regulatory approvals for such project types to prove that they are in compliance with relevant laws, statutes and regulatory frameworks in Turkey are the Environmental Impact Assessment (EIA) and the Electricity Generation License (EGL) whose obtainments are required for projects being in compliance with other relevant legal requirements.

The project has obtained the approval of the Environmental Impact Assessment (EIA) issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on March 01, 2017, and the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey on August 01, 2019.

Hence, the project is in compliance with the laws, statutes, and other regulatory frameworks of the host country, Turkey.

¹⁶ According to the EPIAŞ which constantly monitors the electricity generation in Turkey, fossil fuel-based electricity generation accounts for approximately 55%. <https://seffaflik.epias.com.tr/transparency/> Accessed on 23.06.2022

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project did not apply to and is, hence, not registered under any other approved and/or voluntary GHG programs, such as CDM¹⁷, JI¹⁸, the Climate Action Reserve¹⁹, Gold Standard for Global Goals²⁰ (GS4GG), International Carbon Registry²¹ (ICR), BioCarbon²² and Global Carbon Council²³ (GCC), rather than VERRA/VCS²⁴.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

If yes, provide the name of the emissions trading program or other mechanism that allows GHG allowance trading.

Relevant official declarations prepared by PPs are provided to the VVB in time of validation.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes

☒ No

¹⁷ <https://cdm.unfccc.int/Projects/projsearch.html> (Accessed on 08/12/2023)

¹⁸ https://ji.unfccc.int/JI_Projects/ProjectInfo.html (Accessed on 08/12/2023)

¹⁹ <https://www.climateactionreserve.org/> (Accessed on 08/12/2023)

²⁰ <https://registry.goldstandard.org/projects?q=&page=1> (Accessed on 08/12/2023)

²¹ <https://iceland.itmoregistry.net/Public/Project> (Accessed on 08/12/2023)

²² <https://biocarbonregistry.com/en/projects/> (Accessed on 08/12/2023)

²³ https://projects.globalcarboncouncil.com/pages/submitted_projects (Accessed on 08/12/2023)

²⁴ <https://registry.verra.org/app/search/VCS> (Accessed on 08/12/2023)

If yes, provide the name of the other program(s) under which the project has sought or received another form of GHG-related credit.

Relevant official declarations prepared by PPs are provided to the VVB in time of validation.

Supply Chain (Scope 3) Emissions

Supply Chain (Scope 3) emissions are assessed in line with the VCS Standard 4.5 and Corporate Value Chain (Scope 3) Accounting and Reporting Standard²⁵. According to these sources, this section is not applicable since the project does not directly impact emissions associated with any good or service other than the electricity supplied to the national grid. However, according to the statement given at the Corporate Value Chain (Scope 3) Accounting and Reporting Standard “*The generator does not account for Scope 3 emissions associated with sold electricity because the emissions are already accounted for in Scope 1.*”

Consequently, the project activity is not related with any Scope 3 emissions. Nevertheless, the PP has taken the action to notify manure supplier farm owners regarding the VCU ownership as mentioned in Section 1.12 of this PD.

1.17 Sustainable Development Contributions

The project contributes to five of the UN Sustainable Development Goals (SDGs) in total. These SDGs are given below;

- SDG 7 Affordable & Clean Energy,
- SDG 8 Decent Work & Economic Growth,
- SDG 12 Responsible Consumption & Production
- SDG 13 Climate Action,
- SDG 17 Partnerships for The Goals.

The project has an electricity generation capacity of 2.406 MWe which is partly fed into the Turkish national grid (SDG 7.2.1.).

The project activity has created some permanent jobs for maintenance and operation during the project activity. In addition, the employees were provided with occupational health and safety trainings (SDG 8.0).

The project recycles/recovers significant amount of waste generated at neighboring animal farms by other cattle during the project activity (SDG 12.0).

The project is expected to generate 32,857 tons of CO₂eq emissions reduction in total per annum (SDG 13.0).

²⁵ https://ghgprotocol.org/sites/default/files/standards/Corporate-Value-Chain-Accounting-Reporting-Standard_041613_2.pdf

The project has a contractual agreement with the Turkish state under the Public-Private Partnership-based (PPP) Renewable Energy Support Scheme (YEKDEM) aiming at increase in electricity generation from renewable sources in Turkey. The scheme has enabled the project to receive USD 0.133 per KWh fed to the national grid (SDG 17.17.1).

The project developer is going to complete the VERRA's "Sustainable Development Contributions Report"²⁶ for the project. Hence, the more detailed data on project's SDG Contribution can be found both in the contribution report and the final monitoring report once they are submitted.

1.18 Additional Information Relevant to the Project

Leakage Management

Leakage emissions are not accounted for in line with the applied methodology of AMS-III.D, version 21.0, referred to (Section 4.3 of VCS PD). Leakage emissions from storage of digestate are not accounted for because the storage activity does not include within the project activity. Further, leakage from the composting of digestate is not accounted for since the project activity does not involve composting process.

Therefore, leakage management is not required for this project. Please find further details in Section 4.3 of VCS-PD.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

2 SAFEGUARDS

2.1 No Net Harm

The Environmental Impact Assessment (EIA) Report has been approved and issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on March 01, 2017 (Please see Appendix-4). In accordance with the Turkish laws and regulations, EIA's approval shall be only made if a project subjected to the approval does not make any negative environmental and socio-economic impacts. Considering the fact that the proposed project activity already obtained EIA approval by the Ministry of Environment and Urbanization in Turkey, it is therefore possible to claim that there is no net harm linked to the project.

²⁶ <https://verra.org/sustainable-development-contributions-report-template-now-available/> (Accessed on 08/12/2023)

2.2 Local Stakeholder Consultation

The procedures or methods used for engaging local stakeholders

It was decided by the Project Personnel to organize a local stakeholder consultation meeting to inform the stakeholders about technical and non-technical issues about the facility. In order to determine the meeting date and the topics to be discussed during the meeting, Dr. Önder met with project officials, project consultants and consultation meeting organizers. Considering the cases due to the Covid-19 pandemic in the world, it was decided to hold the meeting online. The meeting date was determined as 06.07.2022, invitation letters were prepared, and stakeholder consultation as per Section of VCS Standard 4. 3²⁵.

The agenda which was shared with the local stakeholders before and during the LSC meeting can be seen below:

Date of the Meeting: 06.07.2022	
Time: 14:00 – 16:00	
Location: Online Webinar, Microsoft Teams	
Agenda:	
14:00 - 14:05	- Opening of the meeting & Introduction
14:05 - 14:20	- Explanation of the project
14:20 - 14:40	- Brief introduction to climate change, sustainable development, carbon offsetting mechanisms and certification processes
14:40 - 15:00	- Q&A Session about the project
15:00 - 15:15	- Discussion of continuous input/grievance mechanism
15:15 - 15:30	- Sustainable Development Contributions (SDC) explanation
15:30 - 16:00	- Discussion on monitoring SD & Closure of the meeting

In order to reach out the locals living in the Elmadağ municipality, as well as in the Ankara province, the project personnel decided to make a public announcement about the LSC meeting in a local newspaper. Personnel made the public announcement, consisting of the LSC meeting's agenda, location, and date, along with a brief project description, in the local newspaper "Sonsöz" on June 29th, 2022.

²⁵https://verra.org/wp-content/uploads/2022/09/VCS-Standard_v4.3.pdf (Accessed on 08/12/2023) Since the most recent VCS Standard was available in time of Local Stakeholder Consultation, VCS Standard v4.3 was considered.

The published announcement in the Sonsöz Newspaper dated June 29th, 2022, is as follows:

[illegible]

During a two-day long site-work (June 27th and 30th) announcements were carried out to reach the identified stakeholders. Project Personnel visited the Mukhtar to the invite to the LSC meeting and hanged the invitation letter on the villages' announcement boards and/or the Mukhtars' offices. Also the invitation letters were posted/delivered by hand in the Mukhtar Office of Kurtuluş Neighbourhood, Elmadağ Municipality, Elmadağ District Directorate of Agriculture and Forestry which are the most frequently visited spots in the Elmadağ Municipality, to inform the local people. Three of the samples of hanging the invitations can be seen below.

The places where the invitation letters are hanged are as follows:

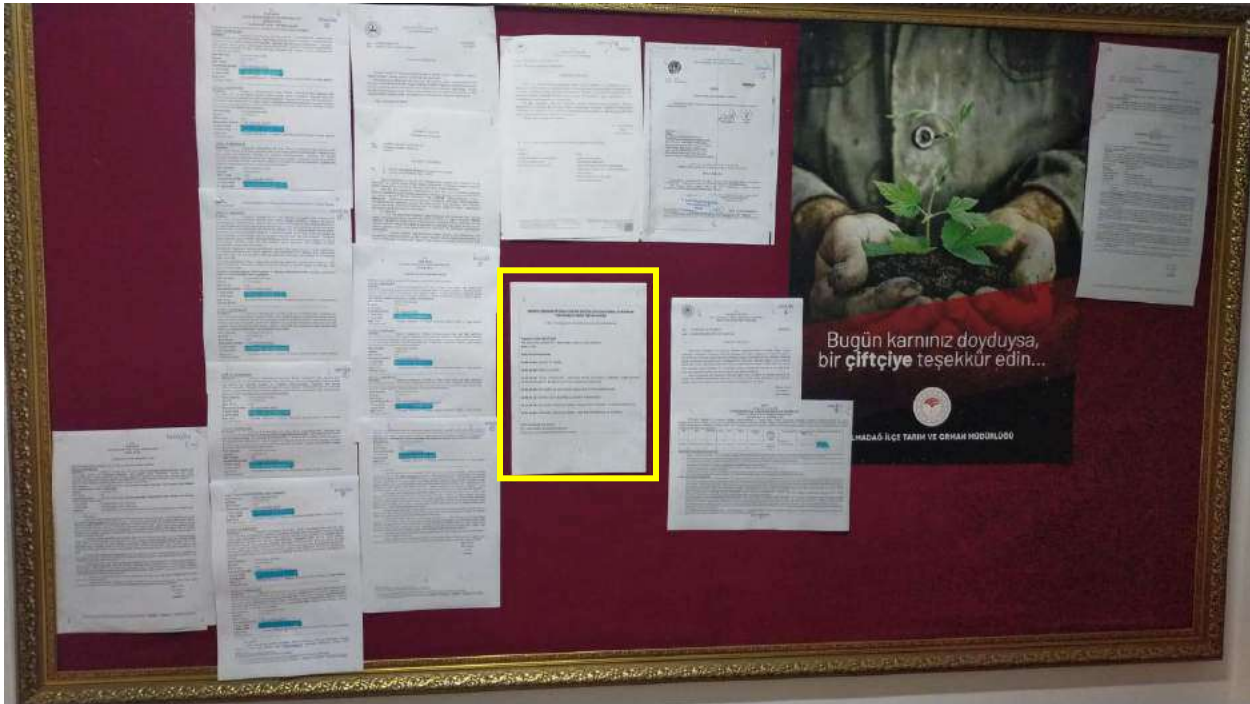


Figure 11. Elmadağ Municipality Information Board



Figure 12. Elmadag Directorate of Agriculture and Forestry Information Board

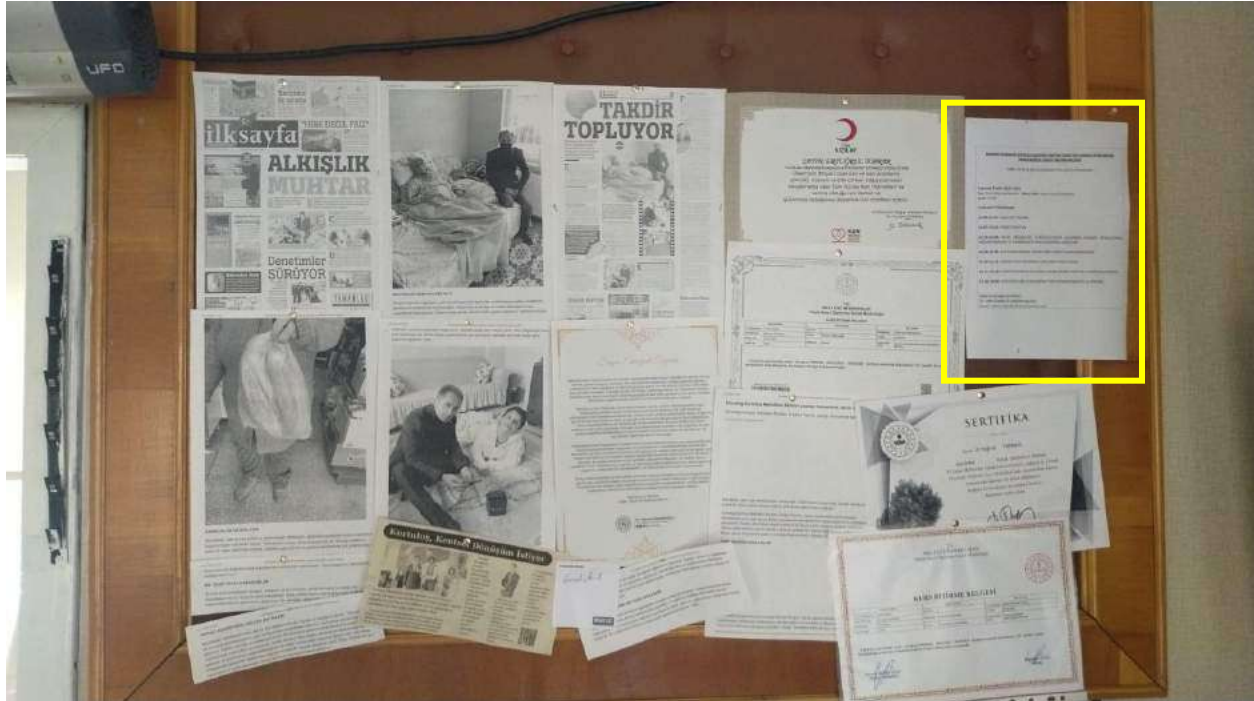


Figure 13. Kurtuluş Neighbourhood Information Board

Project Personnel also delivered the Invitation Letter, the Key Project Information document and the LSC Meeting Agenda to the Mukhtar. This "hand-delivered" document with wet signatures from the Mukhtar can be seen below:



Figure 14. Hand-delivered document of Kurtuluş Neighborhood Mukhtar

Due to the increase in Covid-19 cases in and around the relevant province, meeting invitations were made by sending e-mails and sending messages to the relevant institutions, organizations, non-governmental organizations, and individuals by avoiding hand delivery. Due to the constant communication of the project authorities, national authorities, regional and provincial administrations were personally called and invited.

The invitation tracking table is as follows:

Category code	Organisation (if relevant)	Method of invitation	Date of invitation
1	Local people from Ankara Province	Public announcement published in the local newspaper: Sonsöz	29 June 2022
1	People of Kurtuluş Neighborhood	Printed invitation hanged at the Mukhtar's office and the Mukhtar received the documents in person	27 June 2022

2	Republic of Turkey Ministry of Environment, Urbanization and Climate Change	The invitation documents were sent by email	30 June 2022
2	General Directorate of State Hydraulic Works	The invitation documents were sent by email	30 June 2022
3	Ankara Municipality	The Invitation documents hand delivered	30 June 2022
3	Ankara Environmental Protection and Control Department	The Invitation documents hand delivered	30 June 2022
4	WWF	The invitation documents were sent by email	30 June 2022
4	Greenpeace	The invitation documents were sent by email	30 June 2022

Stakeholder invitations were provided via e-mail and whatsapp messages by communicating electronically due to the Covid-19 pandemic experienced in the world and in our country. Invitations to the meeting were hand-delivered to the Governorship of Ankara, with which the project is in close contact. In addition, invitation letters, verbal invitations, e-mails, newspaper advertisements and announcements of meeting invitations were posted in local venues to inform the above stakeholders. In addition, in order to facilitate the participation of local people in the online meeting, registration and participation methods are explained in detail to the local people in the places where announcements are posted.

Minutes of the LSC Meeting

The local stakeholder information meeting was held on 6th July 2022 as agreed. At the beginning of the meeting, it was explained in detail to the participants of the online meeting, which was held due to the Covid-19 outbreak, how questions about the project would be collected during the meeting and how they could give feedback.

The demographic information of the participants was received during the registration for the online meeting. On the registration page of the meeting held over Microsoft Teams, non-technical information about the project is also summarized and explained.

The English translations of the personal information received at the registration page of the stakeholder consultation meeting are respectively: 'Name'; 'Surname'; 'E-mail address'; 'Gender'; 'Age'; 'Job Title' and 'City and District of Residence'

Registration page is viewed 88 times and total number of registrants to the meeting is 21.

Microsoft Teams, Local Stakeholder Consultation Webinar registration page:

ANKARA ELMADAĞ BİYOGAZ ELEKTRİK ÜRETİM TESİSİ HAYVANSAL ATIKLARDAN YENİLENEBİLİR ENERJİ ÜRETİM PROJESİ YEREL PAYDAŞ BİLGİLENDİRME TOPLANTISI PROGRAMI

Etkinlik ayrıntıları

06.07.2022 14:00 - 06.07.2022 16:00
Saat dilimi: (GMT+03:00) İstanbul

Microsoft Teams toplantısı

ANKARA ELMADAĞ BİYOGAZ ELEKTRİK ÜRETİM TESİSİ
HAYVANSAL ATIKLARDAN YENİLENEBİLİR ENERJİ ÜRETİM
PROJESİ

YEREL PAYDAŞ BİLGİLENDİRME TOPLANTISI PROGRAMI

Abrak Enerji Elektrik Üretim A.Ş. olarak Ankara İli, Elmadag İlçesi sınırları içerisinde işlettiğimiz biyogaz tesisinde; küresel iklim krizi ile mücadelede, yenilenebilir enerji üretiminin artmasına, enerji tüketiminde dışa bağımlılığının azaltılmasına katkı sağlamayı, hayvansal atıkları değerlendirerek yenilenebilir enerji üretmeyi ve bölgede oluşan çevresel etkileri azaltmayı hedeflemekteyiz.

Firmamız, projenin yaratmış olduğu sera gazı emisyon azaltımının uluslararası seviyede belgelendirilmesi amacıyla VCS Gönüllü Emisyon Azaltım sertifikalandırma sürecini başlatmış bulunmaktadır.

Bu kapsamda değerli görüşlerinizi almak ve sizleri bilgilendirmek için düzenlenecek olan, Yerel Paydaş Bilgilendirme Toplantısı'nda siz halkımız, kurum ve kuruluşları bu etkinlikte görmekten memnuniyet duyacağımızı belirtmek isteriz.

Toplantıya katılımlarınızın projemiz için önemli olduğunu tekrar belirtir, konu ile ilgili olarak bilgi ve gereğini arz ederiz.

Toplantıya katılımlarınızın projemiz için önemli olduğunu tekrar belirtir, konu ile ilgili olarak bilgi ve gereğini arz ederiz.

Toplantı Tarihi: 06.07.2022
Toplantı Yeri: Çevrimiçi yapılacaktır.
Toplantı Saati: 14:00
Kayıt Linki: <https://t.ly/ELMADAG>

TOPLANTI PROGRAMI:
14:00-14:05: AÇILIŞ VE TAKDİM
14:05-14:20: PROJE TANITIMI
14:20-14:40: İKLİM DEĞİŞİKLİĞİ SÜRDÜRÜLEBİLİR KALKINMA, KARBON DENKLEŞTİRME
MEKANİZMALARI VE KARBON SERTİFİKALANDIRMA SÜREÇLERİ
14:40-15:00: KATILIMCILARDAN GELEN SORULARIN CEVAPLANDIRILMASI
15:00-15:15: SÜREKLİ GERİ BİLDİRİM SÜRECİNİN TANITILMASI
15:15-15:30: SÜRDÜRÜLEBİLİR KALKINMA ADIMLARININ TANITIMI ve DEĞERLENDİRMESİ
15:30-16:00: SÜRDÜRÜLEBİLİR KALKINMA TAKİP MEKANİZMALARI ve KAPANIŞ

Toplantıya katılamayacak olup görüşlerini bildirmek isteyen ilgililerin umut.onder@climatesolutions.com adresi ile iletişime geçmesini rica ederiz.

Bu etkinliğe kaydol

* Gerekli

Ad *

Soyadı *

E-posta *

Yaş *

Cinsiyet *

☐ Kadın

☐ Erkek

İş unvanı *

İkamet edilen il ve ilçe *

Şimdi kaydolun

Microsoft Gizlilik Bildirimi.

Meeting participation list is exported from Microsoft Teams. The participants who registered to the meeting are as follows:

Name Surname	Age	Gender	Job Title	City and District of Residence	Registration Date
Refiye Almaz	29	Female	chemical engineer	ANKARA/ELMADAĞ	2.07.2022 11:31
İSMAIL Bostan	45	Male	electrical technician	elmadağ	2.07.2022 12:36
Burak Bilgi	23	Male	Logistics/Weighbridge	Ankara/Altındağ	2.07.2022 13:59
Samet Ay	27	Male	energy engineer	Hasanoglan	4.07.2022 13:06
Ali Büken	28	Male	Mechanical engineer	Türkiye	4.07.2022 13:07
baran salttürk	25	Male	automation	şehitlik mahallesi harman sokak no 27	6.07.2022 08:30
Ömer Kaan KURUSAKIZ	28	Male	City planner	Ankara/Elmadağ	6.07.2022 08:47
burhan kement	29	Male	R&D Specialist	Ankara Elmadağ	6.07.2022 08:49
Necati Ateşci	29	Male	R&D Specialist	Ankara - Keçiören	6.07.2022 09:02
zeki ışık	28	Male	ENGINEER	ANKARA MAMAK	6.07.2022 09:24
Fatih Furkan Hatipoğlu	30	Male	Engineer	Ordu/Ünye	6.07.2022 09:33
Seyit Ali Küçükkahraman	54	Male	Mukhtar	Elmadağ /Ankara Elmadaengiz	6.07.2022 10:37
Selçuk ZORGEÇEN	37	Male	Accountant	Ankara/Elmadağ	6.07.2022 10:46
Nadir Çetin	56	Male	Mukhtar	Elmadağ HASANOĞLAN Muzaffer Ekşi Mahallesi	6.07.2022 11:01
Mertcan Akdeniz	27	Male	Construction engineer	Antalya / Alanya	6.07.2022 11:07
MUHSİN YİĞİT YAĞCI	36	Male	Teacher	Ankara Elmadağ	6.07.2022 11:09
Ertuğrul Toprak	43	Male	Mukhtar	Ankara/Elmadağ	6.07.2022 12:31
Mehmet GÜVEN	64	Male	retired	Elmadağ-Ankara	6.07.2022 12:45
Mert Çeneli	35	Male	environmental engineer	Ankara	6.07.2022 13:44
BÜLENT HASAN	64	Male	retired	ankara keçiören	6.07.2022 13:57
Tuncay Kara	48	Male	retired	Elmadağ	6.07.2022 14:51

(a) Opening of the meeting & Introduction (14:00-14:05):

The person appointed as the Moderator of the meeting greeted the participants and shared the purpose and agenda of the meeting. Then, chemical engineer of the project Mrs. Refiye Almaz, who is actively working in the project and its field, was invited to present the technical description of the project.

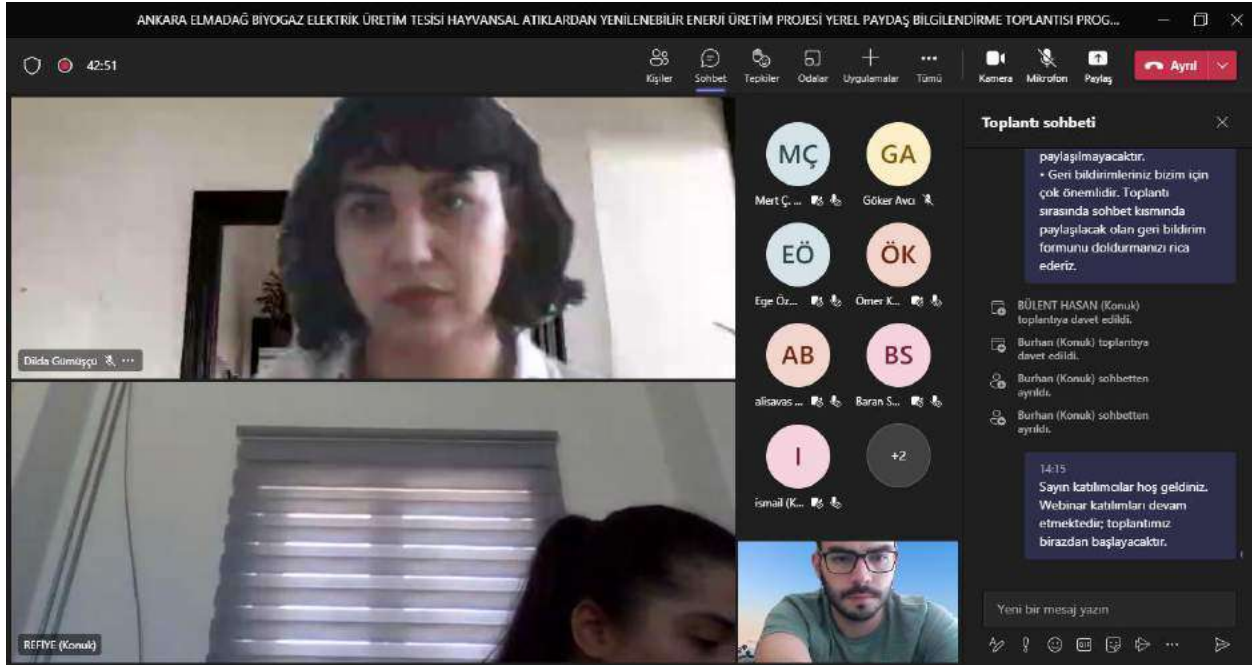


Figure 15. Opening of the meeting & Introduction

(b) Project Introduction and Description (14:05-14:20):

The project chemical engineer Mrs. Refiye Almaz made a presentation of the improvements at the project site. They provided detailed information on how the Elmadag Biogas Energy Facility functions and talked about the technical information in a way that could easily be understood by the audience. Her presentation included the following points:

- Background of the project site;
- What Biogas is; - How a biogas facility would generate renewable energy;
- Waste characterization of the incoming waste stream;
- Technical details of the;
- The installed capacity of the facility;
- The main motivation behind implementing the Project;
- Its exact location;
- Its main outcomes in terms electricity;
- Its process flow chart;
- Its main components;
- Its main phases;
- The timelines for implementing the Project;

- The key milestones of the Project;

ANKARA ELMADAĞ BIYOGAZ ELEKTRİK ÜRETİM TESİSİ HAYVANSAL ATIKLARDAN YENİLENEBİLİR ENERJİ ÜRETİM PROJESİ YEREL PAYDAŞ BİLGİLENDİRME TOPLANTISI PROJESİ...

Denetim iste

Kişiler Sohbet Tepkiler Odalar Uygulamalar Tüm

Kamera Mikrofon Paylaş

Ayrı

Toplantı sohbeti

14:17
Sayın katılımcılar, projeye dair sorularınız için aşağıdaki linke tıklayarak sorularınızı iletmenizi rica ederiz.
<https://forms.office.com/r/7D10aiw7hQ>

Görüşlerinizi ve sorularınızı ayrıca belirtmek isteyen ilgililer için umut.onder@climatesolutions.com adresi ile iletişime geçmenizi rica ederiz.

14:21
Sayın katılımcılar, projeye dair sorularınız için aşağıdaki linke tıklayarak sorularınızı iletmenizi rica ederiz.
<https://forms.office.com/r/7D10aiw7hQ>

Yeni bir mesaj yazın

KOJENERASYON

ENERJİNİN HEM ELEKTRİK HEM DE ISI FORMLARINDA AYNI SİSTEMDEN ÜRETİLMESİDİR VE İKİ ENERJİ FORMUNUN DA TEK TEK KENDİ BAŞLARINA AYRI YERLERDE ÜRETİLMESİNDEN DAHA EKONOMİKTİR.

Kojenerasyon Üniteleri

Santraldeki işlemler sırasıyla şu şekilde gelişir;

- 1.Kompresör basınçlı havayı yakma hücresine gönderir.
- 2.Yakıt kaynağından, yakma hücresine yakıt gelir.
- 3.Yanma ürünü sıcak gazlar, gaz türbinine gelerek rotasyonel hareketi sağlar.
- 4.Rotasyonel enerji, jeneratör aracılığıyla elektrik enerjisine çevrilir.
- 5.Buraya kadar olan işlemler, geleneksel elektrik üretiminde kullanılan adımlardır. Bu noktadan itibaren, kojenerasyon teknolojisi başlar. Önceki adımda yanma ürünü gazlar, atık ısı değerlendirme kazanına gelerek buhar üretiminde kullanılır.
- 6.Oluşan buharın bir kısmı, ısıtma ve diğer ihtiyaçlarında kullanılır.
- 7.Diğer kısmı ise buhar türbinine hareketi sağlar.
- 8.Rotasyonel enerji, jeneratör aracılığıyla elektrik enerjisine çevrilir.

REFİYE (Konuk)

Figure 16. Project Introduction and Description by Chemical Engineer Refiye Almaz

The rules of the webinar, question submission links and feedback form links about the project were continuously shared by the moderator in the "chat" section. Also, Dr. Önder's mail address was continuously shared as a part of continuous feedback mechanism.

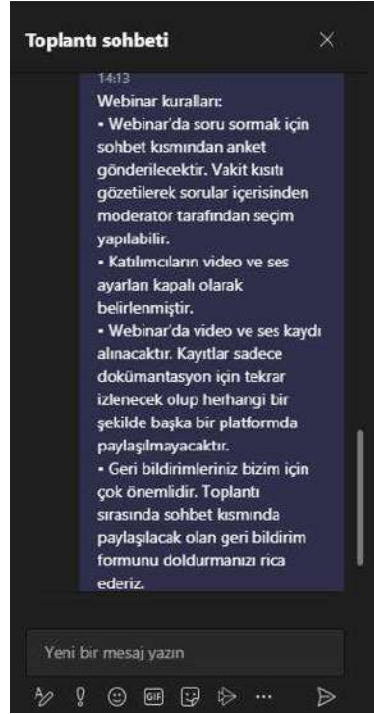


Figure 17. Webinar Rules

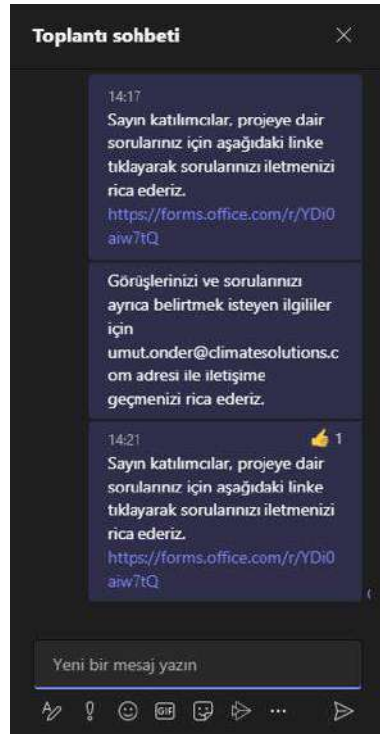


Figure 18. Link for attendees questions

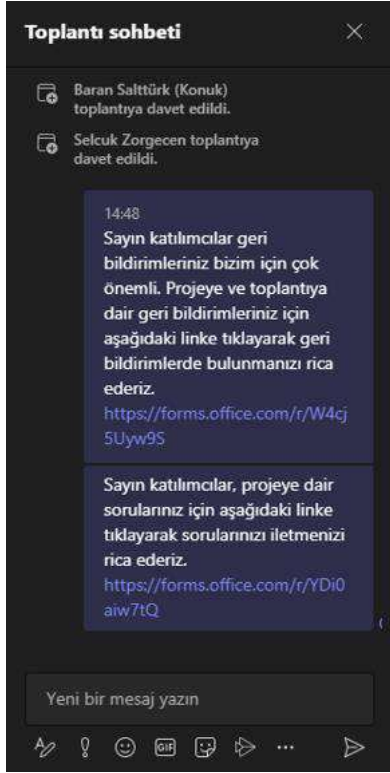


Figure 19. Links for questions & meeting feedbacks

(c) Brief introduction to climate change, sustainable development, carbon offsetting mechanisms and registration processes (14:20-14:40):

Climate & Sustainability Consultant Mrs. Yüzer, provided insights into the historical backgrounds of Sustainable Development and Climate Change at the international arena, including, the establishment of UNFCCC in 1994 and the 1997 Kyoto Protocol. Then, he briefly explained what the UN's Sustainable Development Goals were. During her presentation, the audience were informed about how they could learn more about the SDGs by showing the official UN's web-sites in Turkish, the local language. She continued by explaining main greenhouses gas types and results of this gases on the world and gave an end to her speech by providing insight about how to manage a biogas site in an environmentally friendly manner.

Mr. Göker Avcı started his speech later on by explaining the audience that one of the central subjects and aims in organizing the meeting was to assess the Project's contribution in achieving sustainable development goals and in tackling climate crisis. He sketchily explained what VERRA/VCS was, how it functioned and what the main requirements were for projects like Ankara Elmadag Biogas Energy Facility to access to its certification.

He continued by explaining MRV – Monitoring, Measuring, Reporting and Verifying processes. He then clarified the two main requirements which a carbon offsetting activity needs to fulfil: 'contribution to sustainable development in host

country/communities' and 'additionality'. While doing so, in order to simplify these complex terms to an appropriate level that met the stakeholders' understanding,

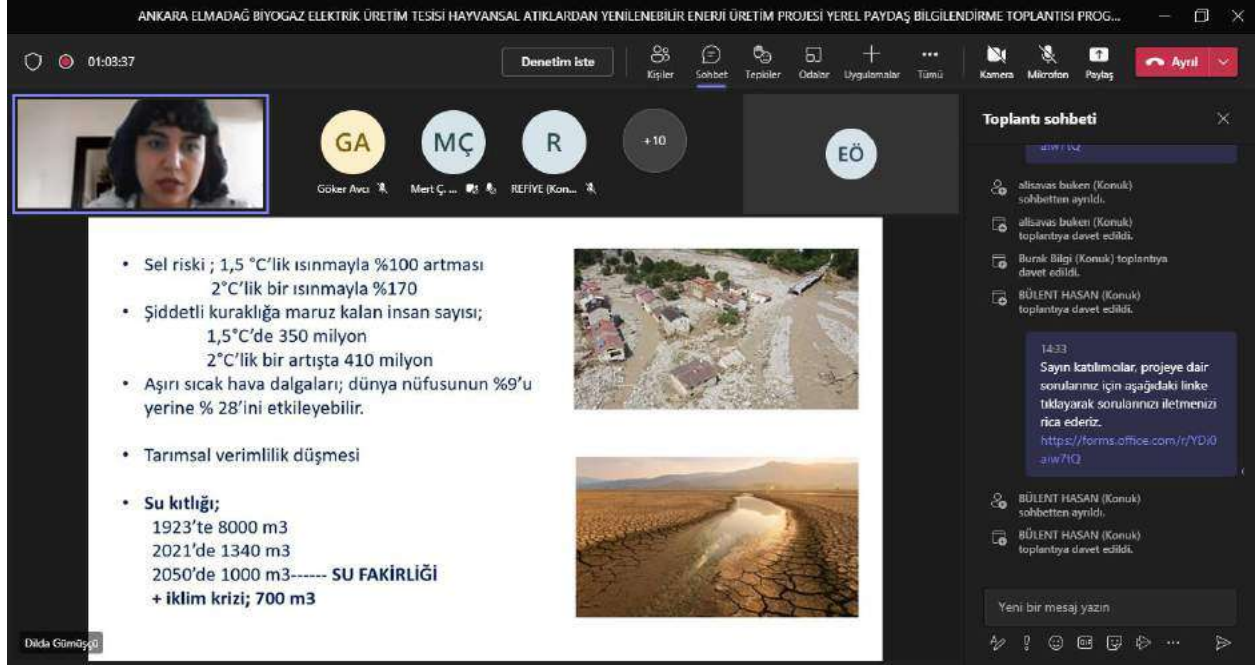


Figure 20. Brief introduction to climate change, sustainable development by Climate & Sustainability Consultant Mrs. Yüzer

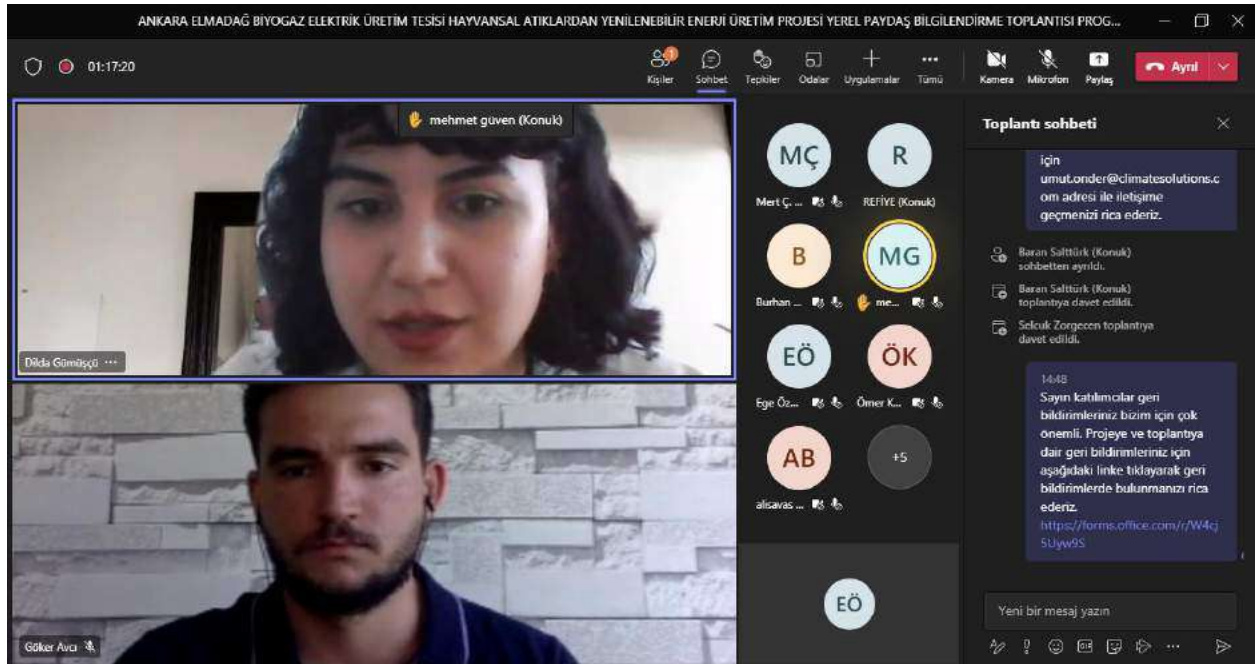


Figure 21. Brief introduction to carbon offsetting mechanisms and registration processes by Mr. Göker Avcı

(d) Questions and Answers (14:40-15:00):

Majority of the questions was linked to the technical details of the project. The questions asked by attendees are as follows;

Q: Hello, what is the difference between a biogas plant and a biomass plant?

A: Biogas facilities collect and process animal manure as feedstock and biomass facilities collect wastes including household organic waste and industrial organic wastes as feedstock.

No further action required, project responsible has made the necessary explanation.

Q: What precautions do you take for your employees in terms of occupational health and safety?

A: We provide the necessary occupational health and safety training to our employees periodically, as per the relevant laws. The processes in our facility takes place in a completely closed environment. We also ensure that our employees use personal protective equipment.

No further action required, project responsible has given the necessary information.

Starting at 15:40, moderator of the meeting checked submitted questions and directed those to the relevant presenters in the meeting.

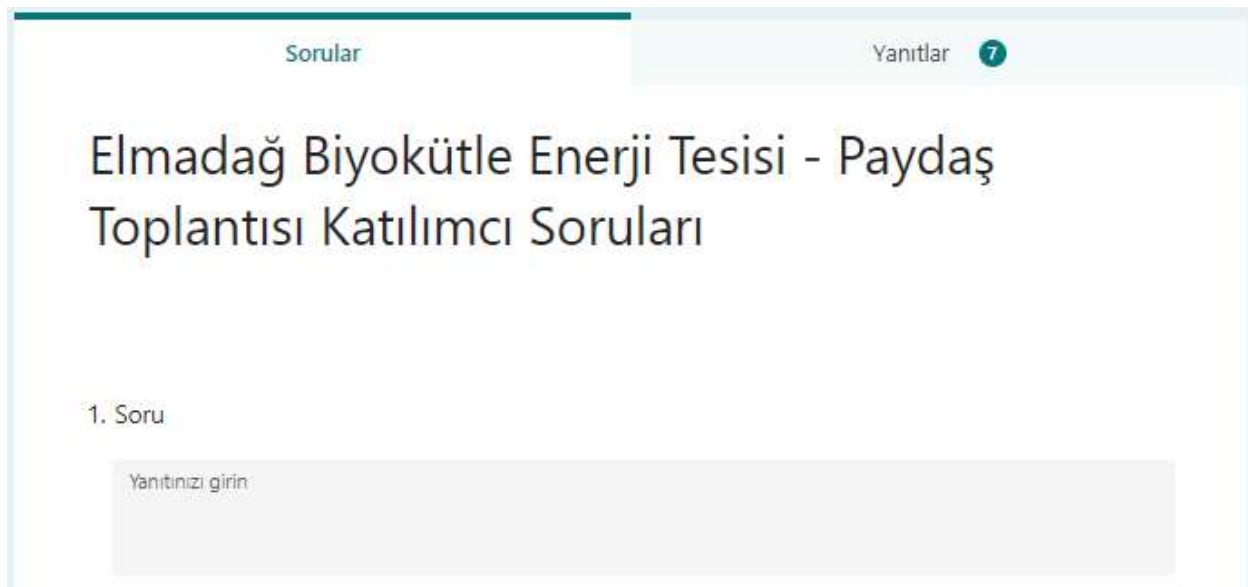


Figure 22. Participants' questioning interface is as above



Figure 23 The questions asked by the participants are as above



Figure 24. Answering incoming questions

(e) Discussion of continuous input/grievance mechanism (15:00-15:15):

With the links shared by the moderator during the webinar, feedback on the project and meeting started to be received. Stating that during the project, feedback forms were left at the security booth at the project site entrance so that the stakeholders could give feedback, and project consultant Dr. Umut Önder's e-mail information (Figure 18) has been shared continuously. For stakeholders who could not participated, e-mail information of Dr Umut Önder also has been shared in invitation letters as shown. In addition, the participants were informed that the project will be published on the VCS website for 1 month and that they can give feedback on the VCS website during this time.

All comments and complaints, which are sent to headman of the nearest villages are responded to either verbally or in writing in accordance with the preferred method of communication specified by the complainant. All grievances are registered and responded within legally approximately in 10 days. If necessary, a meeting with the Company management is organized and will ensure more precise process of collection of grievances.

(f) Discussion on Monitoring Sustainable Development & Closure of the Meeting (15:15-15:45):

At the final section, Sustainable Development Contributions of the project and the monitoring method were presented by the moderator, Mrs. Göker Avcı.

The audience were invited to make comments about the Project's sustainable development impacts. The audience was reminded the Project's proposed positive impacts on the sustainable development, which were discussed earlier at the meeting during the climate crisis information section. The moderator ensured that the audience was aware that they, as the stakeholders, had responsibilities to monitor these impacts. He reminded the stakeholders that their voices would be heard, and their concerns would be considered in every single phase of the proposed Project activity, particularly via the Continuous Input and Grievance Mechanisms explained earlier.

Following the discussion on the Project's impacts on sustainable development, the audience was invited to complete the Local Stakeholder Consultation Meeting Feedback Form. 3 out of 12 attendees completed the forms which were shared on the chat.

Sorular

Yanıtlar 3

Elmadağ Biyokütle Enerji Tesisi - Katılımcı Görüşleri

1

İsim-Soyisim *

Yanıtınızı girin

2

Proje hakkında yeterli bilgilendirme yapıldı mı? *

☐ Evet

☐ Hayır

3

Yapılması planlanan projenin iyileştirme sağlayacağını düşünüyor musunuz? *

☐ Evet

☐ Hayır

☐ Bilmiyorum

4

Yapılması planlanan projenin istihdam oranında bir iyileştirme sağlayacağını düşünüyor musunuz? *

☐ Evet

☐ Hayır

☐ Bilmiyorum

5

Projenin sizin ve ailenize nasıl bir etkisi olabileceğini düşünüyorsunuz? *

- ☐ Pozitif (Olumlu Etki)
- ☐ Negatif (Olumsuz Etki)
- ☐ Etkisi Yok

6

Projenin yerel halk üzerindeki etkisinin ne olacağını düşünüyorsunuz? *

- ☐ Ekonomik kalkınmayı destekler
- ☐ Elektrikçe erişimde iyileştirmeler sağlar
- ☐ Atık Yönetim sistemlerini geliştirir
- ☐ Bilmiyorum
- ☐ Diğer

7

17 Küresel Hedef ile değerlendirildiğinde projenin pozitif ilişkisi olduğunu düşündüğünüz en az üç hedefi seçiniz. *



- ☐ Yoksulluğa Son
- ☐ Açlığa Son
- ☐ Sağlıklı Bireyler



☐ Nitelikli Eğitim
☐ Toplumsal Cinsiyet Eşitliği
☐ Temiz Su ve Sıhhi Koşullar
☐ Erişilebilirlik ve Temiz Enerji
☐ İnsana Yakışır İş ve Ekonomik Büyüme
☐ Sanayi, Yenilikçilik ve Altyapı
☐ Eşitsizliklerin Azaltılması
☐ Sürdürülebilir Şehir ve Yaşam Alanları
☐ Sorumlu Tüketim ve Üretim
☐ İklim Eylemi
☐ Sudaki Yaşam
☐ Karasal Yaşam
☐ Barış, Adalet ve Güçlü Kurumlar
☐ Hedefler İçin Ortaklıklar
☐ Bilmiyorum

Figure 25. The feedback survey is as above

The answer list of the feedback forms which were completed by attendees during the meeting can be seen below:

ID	Start time	Completion time	Name-Surname	Has enough information been given about the project?	Do you think the planned project will provide improvement?	Do you think that the planned project will provide an improvement in the employment rate?	What effect do you think the project might have on you and your family?	What do you think will be the impact of the project on local people?	Select at least three goals that you think have a positive relationship with the project when evaluated with the 17 Global Goals.
1	7.06.2022 15:00	7.06.2022 15:01	Ali savaş büken	Yes	Yes	I don't know	Positive (Positive Impact)	Supports economic development	Affordable and Clean Energy;Clean Water and Sanitation;Partnerships to achieve the Goal
2	7.06.2022 15:03	7.06.2022 15:04	Samet Ay	Yes	Yes	Yes	Positive (Positive Impact)	Supports economic development	Decent Work and Economic Growth;Affordable and Clean Energy;Reduced Inequality;Climate Action
3	7.06.2022 15:04	7.06.2022 15:05	Baran SALTÜRK	Yes	Yes	Yes	Positive (Positive Impact)	Develops Waste Management systems	Good Health and Well-being;Quality Education;Gender Equality;Clean Water and Sanitation;Sustainable Cities and Communities;Climate Action;Life on Land

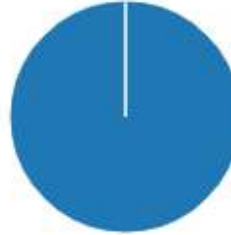
(g) Project & Meeting Feedbacks in Graphs



Question 1 & Answers - Name - Surname

2. Proje hakkında yeterli bilgilendirme yapıldı mı?

[Diğer Ayrıntılar](#)



Question 2 & Answers - Has enough information been given about the project? (Yes – No)

Attendees who gave feedback for the meeting and project said "yes" to this question. Therefore no further action required.

3. Yapılması planlanan projenin iyileştirme sağlayacağını düşünüyor musunuz?

[Diğer Ayrıntılar](#)



Question 3 & Answers - Do you think the planned developments will provide improvement?

(Yes – No – I don't know)

Attendees who gave feedback for the meeting and project said "yes" to this question. Therefore no further action required.

4. Yapılması planlanan projenin istihdam oranında bir iyileştirme sağlayacağını düşünüyor musunuz?

[Diğer Ayrıntılar](#)

● Evet	3
● Hayır	0
● Bilmiyorum	0



Question 4 & Answers - Do you think the project will provide an improvement in the employment capabilities of the local area? (Yes – No – I don't know)

Attendees who gave feedback for the meeting and project said "yes" to this question. Therefore no further action required.

5. Projenin sizin ve ailenize nasıl bir etkisi olabileceğini düşünüyorsunuz

[Diğer Ayrıntılar](#)

● Pozitif (Olumlu Etki)	3
● Negatif (Olumsuz Etki)	0
● Etkisi Yok	0



Question 5 & Answers - How will the project affect you and your family? (Positive – Negative – No effect)

Attendees who gave feedback for the meeting and project said "positive" to this question Therefore no further action required.

6. Projenin yerel halk üzerindeki etkisinin ne olacağını düşünüyorsunuz?

[Diğer Ayrıntılar](#)

● Ekonomik kalkınmayı destekler	2
● Elektriğe erişimde iyileştirmeler	0
● Atık Yönetim sistemlerini geliştirir	1
● Bilmiyorum	0
● Diğer	0



Question 6 & Answers – How will the project affect local residents?

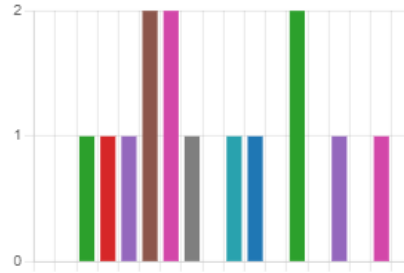
Answers respectively; Supports economic development, improves electricity supply, improves waste management systems, I don't know, other

Respondents answered positively to the question, therefore no further action required.

7. 17 Küresel Hedef ile değerlendirildiğinde projenin pozitif ilişkisi olduğunu düşündüğünüz en az üç hedefi seçiniz.

[Diğer Ayrıntılar](#)

Yoksulluğa Son	0
Açlığa Son	0
Sağlıklı Bireyler	1
Nitelikli Eğitim	1
Toplumsal Cinsiyet Eşitliği	1
Temiz Su ve Sıhhi Koşullar	2
Erişilebilirlik ve Temiz Enerji	2
İnsana Yakışır İş ve Ekonomik Büyüme	1
Sanayi, Yenilikçilik ve Altyapı	0
Eşitsizliklerin Azaltılması	1
Sürdürülebilir Şehir ve Yaşam Alanları	1
Sorumlu Tüketim ve Üretim	0
İklim Eylemi	2
Sudaki Yaşam	0
Karasal Yaşam	1
Barış, Adalet ve Güçlü Kurumlar	0
Hedefler İçin Ortaklıklar	1
Bilmiyorum	0



Question 7 & Answers Pick at least 3 SDG's that you think are the most related to the project.

of Votes per SDG - Good Health and Well-being: 1, Quality Education: 1, Gender Equality: 1, Clean Water and Sanitation: 2, Affordable and Clean Energy: 2, Decent Work and Economic Growth: 1, Reduced Inequality: 1, Sustainable Cities and Communities: 1, Climate Action: 2, Life on Land: 1, Partnerships to achieve the Goal: 1.

In addition, information about the SDGs was also given at the meeting. Participants who gave feedback marked the SDGs which they thought the project was related.

2.3 Environmental Impact

The environmental matters in Turkey are under the jurisdiction of the Ministry of Environment, Urbanization and Climate Change. The Law that defines among others the general requirements of the Environmental Impact Assessment (EIA) states specific requirements according to the type of project, location, size, and other characteristics. The EIA certificate has been approved and issued by the Ministry of Environment and Urbanization in Turkey on March 01, 2017. In accordance with the Turkish laws and regulations, EIA's approval can only be

received if the project does not cause any negative environmental and socio-economic impacts. Thus, the EIA approval by the Ministry of Environment, Urbanization and Climate Change in Turkey clearly demonstrates that there is no net harm linked to the project.

2.4 Public Comments

This project was open for public comment from 10/11/2022 to 10/12/2022, on its official VERRA/VCS webpage²⁷. No comments had been received during the public comment period.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The project uses two approved baseline and monitoring methodologies. The related methodologies are as follows:

- AMS-III.D.: “Methane recovery in animal manure management systems”, Version 21.0²⁸
- AMS-I.D.: “Grid connected renewable electricity generation”, Version 18.0²⁹

Applied tools:

- “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”, Version 03.0³⁰;
- “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, Version 03.0³¹;
- “Project emissions from flaring”, Version 04.0³²;
- “Project and leakage emissions from anaerobic digesters”, Version 02.0³³;
- “Tool to calculate the emission factor for an electricity system”, Version 07.0³⁴;

²⁷ <https://registry.terra.org/app/projectDetail/VCS/3890> (Accessed on 08/12/2023)

²⁸ <https://cdm.unfccc.int/methodologies/DB/H9DVS2407GEZQYLYNWUX23YS6G4RCv> (Accessed on 08/12/2023)

²⁹ <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK> (Accessed on 08/12/2023)

³⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf> (Accessed on 08/12/2023)

³¹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf> (Accessed on 08/12/2023)

³² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-06-v4.0.pdf> (Accessed on 08/12/2023)

³³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-14-v2.pdf> (Accessed on 08/12/2023)

³⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf> (Accessed on 08/12/2023)

3.2 Applicability of Methodology

The AMS-III.D “Approved baseline and monitoring methodology – Methane recovery in animal manure management system.” Version 21.0³⁵ applies to project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant.

The project activity consists of a cattle manure based renewable energy power plant which modifies the existing anaerobic manure management in livestock farms. The manure is collected from several farms to a centralized plant where the project achieves methane recovery, and the recovered methane is used gainfully to generate electricity. As such the project activity satisfies all the three conditions and is a Type III.D small-scale CDM project activity.

The methodology is applicable under the following conditions:

Applicability of AMS-III.D. – v21.0			
Ref.		Application Criteria	Applicability
The methodology is applicable under the following conditions:			
1.	(a)	The livestock population in the farm is managed under confined conditions;	The Project's primary feedstock, cattle-manure, is supplied by animal farms. These farms function under the control of the Ministry of Agriculture and Forestry in Turkey. The farm owners manage the livestock population in their farms in accordance with the relevant Turkish laws and regulations confining conditions, especially regarding animal health, such as Law No. 599636, Law No. 328537 and

³⁵ <https://cdm.unfccc.int/methodologies/DB/H9DVSB24O7GEZQYLYNWUX23YS6G4RCv> (Accessed on 08/12/2023)

³⁶ https://www.tarimorman.gov.tr/Belgeler/ENG/Legislation/law_veterinary_services.pdf (Accessed on 08/12/2023)

³⁷ http://www.zmo.org.tr/mevzuat/mevzuat_detay.php?kod=55 (Accessed on 08/12/2023)

			Regulation on Livestock Farms ³⁸ . These farms are constantly inspected by the relevant governmental institutions to check whether they are managed under confined conditions set by the relevant Turkish laws and regulations. As such, this project activity qualifies as a Type III.D small-scale CDM project activity.
	(b)	Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise "AMS-III.H Methane recovery in wastewater treatment" shall be applied;	As shown in the Section 1.11. of the PD, the waste streams after the treatment will not be discharged into natural water resources. The refined and processed fermented product is sent to the separator and decanter (dehydration unit) to be separated. Both the liquid and solid parts of the residual waste from the digestion is further used as fermented product. Hence, the proposed project activity complies with this criteria/condition. In addition, both the biogas power plant and the farms from where manure is supplied to the plant are constantly inspected by the relevant governmental institutions to check whether they are in compliance with the environmental laws and regulations which also cover dischargement of waste into natural water sources.
	(c)	The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5 °C ³⁹ ;	The annual average temperature of baseline site where manure is collected from multiple farms in ANKARA is 12.0 °C ⁴⁰ . Hence, the proposed project

³⁸ <http://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm> (Accessed on 08/12/2023)

³⁹ The AM0073 methodology mentions (page 1/59), "If monthly average temperature in a particular month is less than 5°C this month is not included in the estimations, as it is assumed that no anaerobic activity occurs below such temperature". The project proponents also applies the requirements of this statement.

			activity complies with this criteria/condition.
	(d)	In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m;	In the baseline scenario, cattle manure generated at farms is stored generally in anaerobic lagoons which have depths over 1 m. The manure is stored in these lagoons for over one month. Hence, the project activity complies with these two criteria. The lagoon depths and retention times are recorded via baseline surveys which will be shared with the VVB in time of validation.
	(e)	No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario	The baseline scenario consists of uncovered anaerobic lagoons that cause methane emissions. Since there is no legal regulation to mandate the livestock and poultry farm owners to implement different anaerobic lagoon designs and to capture and/or utilize methane generated at these lagoons, the continuum of this practice is the most economic, viable and reasonable option for farm owners. Similar to the previous criteria above, in time of validation, results of baseline surveys will be provided as documentary evidence to confirm that no methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario.
2.	(a)	The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of “AMS-III.AO	The residual waste from the project activity is handled aerobically utilized as organic fertilizer which will not result in methane emissions.

⁴⁰ <https://mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=ANKARA> (Accessed on 20/10/2022)

		Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured;	
	(b)	Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared;	The project activity utilizes the biogas for electricity generation. A flaring system is also installed to flare the gas in case of emergencies. Hence, the proposed project activity complies with this criteria/condition.
	(c)	The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply	The manure after removal from the animal barns will be transferred into the anaerobic digesters via waste receiving units within same day.
3.		Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion";	In the project activity, methane is not recovered from landfills and hence AMS-III.G is not applicable. Similarly, the project does not involve wastewater treatment, hence AMS-III.H is not applicable. Moreover, the project boundaries do not include composting of animal manure and hence AMS-III.F is not applicable. Further, the project does not involve co-digestion of animal manure and other organic matter and hence AMS-III.AO is not applicable.
4.		Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should	The recovered biogas is used to produce and deliver power to the grid. Thus, AMS-III.D. is applied to the proposed project activity.

	be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component	
5.	New facilities (Greenfield projects) and project activities involving capacity additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies"	The design of this greenfield project activity is limited to 2.4MW which is within the limit of the maximum output capacity equivalent to up to 15MW and hence complies with the related and relevant requirements in the "Guideline: General Guidelines for SSC CDM methodologies, Version 23⁴¹". In addition, the estimated annual emission reductions that will be generated from the implementation of the proposed project activity is 32,857 tCO₂e/y, which is lower than the threshold of 60,000 tCO₂e/y set in the Guideline. Therefore, the proposed project activity complies with the Guideline. In addition, 4.11, Paragraph 37, of the Guideline, "Procedure to determine baseline scenario for Type II and Type II greenfield projects" lays down a step-wise procedure to identify the baseline scenarios. Accordingly, in Section 3.4. of the PD, all the alternatives of manure disposal presently followed by the farms have been presented and the one applicable baseline scenario has been chosen. Hence, alternatives have been included.
6.	The requirements concerning demonstration of the remaining lifetime of	The project is a Greenfield project, does not involve replacement of equipment;

⁴¹ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190916153417994/MethSSC_guid25.pdf (Accessed on 08/12/2023)

	the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies"	therefore, this condition is not applicable to the proposed project activity.
7.	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity	The generation of the project's emission reductions is estimated as 32,857 tCO ₂ e/y, which is lower than the threshold of 60,000 tCO ₂ e/y. Therefore, the proposed project activity complies with this condition.

The AMS-I.D “Approved baseline and monitoring methodology - Grid connected renewable electricity generation”⁴² comprises renewable energy generation units, such as photovoltaic, hydro, tidal/wave, wind, geothermal and renewable biomass:

- (a) Supplying electricity to a national or a regional grid; or
- (b) Supplying electricity to an identifier consumer facility via national/regional grid through a contractual agreement such as wheeling.

The project activity utilizes the biogas collected from the anaerobic digester to generate electricity, and the power is partly fed to the Turkish national grid. Thus, the project is in line with applicability condition illustrated in Table 1 “Scope of AMS-I.D., AMS-I.F. and AMS-I.A. based on project types” in Appendix of AMS-I.D., Version 18.0.

Also, it is important to note that AMS-I.D. is applicable to the proposed project activity since the project falls under the renewable biomass category as per “Clarifications on Definition of Biomass and Consideration of Changes in Carbon Pools due to a CDM Project Activity” reported in the CDM-EB’s 20th Meeting as Annex 8. The Annex clarifies that “biomass also includes gases and liquids recovered from decomposition of non-fossilized and biodegradable organic material.”

The methodology is applicable under the following conditions:

Applicability of AMS-I.D. – v18.0			
Ref.		Application Criteria	Applicability
The methodology is applicable under the following conditions:			
1.	(a)	Install a Greenfield plant;	The proposed project activity is implemented as a Greenfield Power Plant at the site where there was no

⁴² <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK> (Accessed on 08/12/2023)

			renewable energy power plant operating prior to the implementation of the project activity. Hence, the project implies with this condition.
	(b)	Involve a capacity addition in (an) existing plant(s);	N/A
	(c)	Involve a retrofit of (an) existing plant(s);	N/A
	(d)	Involve a rehabilitation of (an) existing plant(s)/unit(s); or	N/A
	(e)	Involve a replacement of (an) existing plant(s)	N/A
2.	Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:		
	(a)	The project activity is implemented in an existing reservoir with no change in the volume of reservoir;	Since the proposed project activity is not a hydro power plant, this condition is not applicable to the project.
	(b)	The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m ² ;	N/A
	(c)	The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m ²	N/A
3.	If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW		The proposed project activity only involves biogas power generation system, without non-renewable components, thus this condition is not applicable for the project.
4.	Combined heat and power (co-generation) systems are not eligible under this category.		The project activity generates electricity from biogas recovered from animal manure via biogas generators/engines/units and is not a combined heat and power system. Although waste heat generated out of biogas

		engines may be used to warm the digesters, it is not a cogeneration project as per the definition of cogeneration in ACM0018, Version 4.0 ⁴³ , i.e. a “Cogeneration plant is a heat and power plant in which at least one heat engine simultaneously generates both heat and power”. Since there is no heat engine installed within the project boundaries, the proposed project activity shall be considered neither as a cogeneration plant nor a project involving co-generation systems.
5.	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units	This project activity is a green-field project activity which does not involve addition of renewable energy generation units at an existing renewable power generation facility. Hence, the proposed project activity complies with this condition.
6.	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW	As a greenfield project, the proposed project activity the project activity does not involve retrofit or replacement of an existing facility. Therefore, this condition is met by the project.
7.	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III	Methane is recovered from the type III component of project activity and the baseline for

⁴³ <https://cdm.unfccc.int/UserManagement/FileStorage/ZI5NXTD6R4AQGOC307L82MYEK9JSHV> (Accessed on 08/12/2023)

	category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored	the type I component of project activity is determined in accordance with AMS-I.D. As noted in the fourth applicability condition of AMS-I.D. above -regarding to be co-generation plant-, the proposed project activity aims to recover and utilize the methane from animal manure in digesters via biogas engines/units/generators to generate electricity which will be fed to the national grid in Turkey. Thus, the baseline for the electricity component shall be in accordance with procedure prescribed under AMS-I.D.
8.	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply	The proposed project activity’s main feedstock is animal manure. Thus, this condition is not applicable to the project.

Apart from the applied Methodologies, the following tools used by the project’s ER calculations are applicable for the project activity.

- Methodological tool: Project Emissions from Flaring, v04.0

Applicability of Project emissions from flaring, v04.0			
Ref.	Application Criteria	Applicability	Justification
1.	Methane is the component with the highest concentration in the flammable residual gas	Applicable	Methane is the component with the highest concentration in the biogas generated at the plant. Hence applicable.

2.	The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g., biogas, landfill gas or wastewater treatment gas).	Applicable	The source of the residual gas in the case of the proposed project activity is biogas. Hence applicable.
3.	The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion.	Applicable	In the proposed project activity, there is no use of auxiliary fuels and the residual gas, as biogas, will be flared in the case of emergency and/or if the gas engines are not functioning. Hence applicable.

- Methodological tool: Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion, v03.0

Applicability of Tool to calculate project or leakage CO ₂ emissions from fossil fuel combustion, v03.0			
Ref.	Application Criteria	Applicability	Justification
1.	The tool is applicable in cases where CO ₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties.	Applicable	In the case of the proposed project activity, CO ₂ emissions from fossil fuel are calculated based on the quantity of fuel combusted by the engines of vehicles transporting manure, as part of the project's emissions. Hence applicable.

- Methodological tool: Tool to calculate the emission factor for an electricity system, v07.0

Applicability of Tool to calculate the emission factor for an electricity system, v07.0			
Ref.	Application Criteria	Applicability	Justification
1.	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline	Applicable	In the case of the proposed project activity, the tool may be applied since the

	emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).		project substitutes grid electricity that is where the project supplies electricity to the Turkish national grid. Hence applicable.
2.	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	Applicable	The project activity is a greenfield grid power plant. All the electricity generated by the project, after part of which is used for auxiliary consumption at the project site, is supplied only to the Turkish National Grid.

3.	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Applicable	Although Turkey is an Annex-I country, Turkey's contribution to global greenhouse gas (GHG) emissions is considerably below the average of Annex-I countries, thus Turkey is officially considered in a different situation from the other Annex-I countries ⁴⁴ . This makes Tool 07 is applicable for the project.
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- Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 03.0

Applicability of Tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 03.0			
Ref.	Application Criteria	Applicability	Justification
1.	If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption: (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at	Applicable	The proposed project activity, the proposed project activity may only purchase electricity from the Turkish national grid and no captive power plant is installed at the site of electricity consumption. The project does not consume electricity from an off-grid fossil fuel fired captive power plant nor a fossil fuel fired captive

⁴⁴ https://unfccc.int/sites/default/files/resource/cp2021_inf02_0.pdf (Accessed on 03/07/2023)

⁴⁴ <https://www.mfa.gov.tr/united-nations-framework-convention-on-climate-change-unfccc-and-the-kyoto-protocol.en.mfa> (Accessed on 02/06/2023) It is stated in this link that "...However, Turkey had no historical responsibility for greenhouse gas emissions in 1992 despite being included in Annex-I. Indeed, by that time, Türkiye was in the last row of the Annex I list among 36 countries in terms of greenhouse gas emission with per capita ratio of 3,88 tCO₂e (less than 1% of the global total), while the average greenhouse gas emission of the Annex I countries was 14,37 tCO₂e per capita."

	the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer; (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer. The captive power plant(s) can provide electricity to the electricity consumer. The captive power plant(s) is/are also connected to the electricity grid. Hence, the electricity consumer can be provided with electricity from the captive power plant(s) and the grid.		power plant is installed at the site of the electricity consumer supplying electricity to the consumer. Hence, Scenario A applies to this project activity. As per the applicability condition, since this project complies with one (Scenario A) of the three Scenarios, this tool is applicable.
2.	This tool can be referred to in methodologies to provide procedures to monitor amount of	Applicable	The proposed project activity supplies the electricity generated at the plant partly to the Turkish

	electricity generated in the project scenario only if (Scenario I) one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.		national grid while utilizing part of the electricity generated for auxiliary (internal) consumption required at the project site. No electricity is supplied to any consumers/electricity consuming facilities. Hence Scenario I apply to this project activity. As per the applicability condition, since this project complies with one (Scenario I) of the three Scenarios, this tool is applicable.
3.	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.	Applicable	There are no captive renewable power generation technologies installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. Only CO₂ emissions are accounted for within the calculations following this tool. Hence applicable.

- Methodological tool: Project and leakage emissions from anaerobic digesters, Version 02.0

Applicability of Tool: Project and leakage emissions from anaerobic digesters			
Ref.	Application Criteria		Applicability
1.	The following sources of project emissions are accounted for in this tool:		
	(a)	CO ₂ emissions from consumption of electricity associated with the operation of the anaerobic digester;	Since the applied methodology only refers to this tool for the calculation of

			leakage emissions, this criterion is not applicable.
	(b)	CO2 emissions from consumption of fossil fuels associated with the operation of the anaerobic digester;	Since the applied methodology only refers to this tool for the calculation of leakage emissions, this criterion is not applicable.
	(c)	CH4 emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester);	Since the applied methodology only refers to this tool for the calculation of leakage emissions, this criterion is not applicable.
	(d)	CH4 emissions from flaring of biogas	Since the applied methodology only refers to this tool for the calculation of leakage emissions, this criterion is not applicable.
2.	The following sources of leakage emissions are accounted for in this tool:		
	(a)	CH4 and N2O emission from composting of digestate;	Since the project activity does not involve composting process this criterion not applicable to the project.
	(b)	CH4 emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond	Since digested from the facility is not sent to a SWDS or kept in a storage or evaporation pond, this criterion is not applicable.
3.	Emission sources associated with N2O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.		Within the scope of this project, emission sources associated with N2O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land

		application of the digestate are neglected because these are minor emission sources.
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Hence, the applied Meth., together with the Tools used in the Meth., are applicable to the proposed project activity.

3.3 Project Boundary

As noted above, the baseline and monitoring methodologies applied to the proposed project activity are AMS-III.D. “Methane recovery in animal manure management systems”, Version 21.0, and AMS-I.D. “Grid connected renewable electricity generation”, Version 18.0.

In accordance with the Methodology, the spatial extent of the project boundary encompasses:

- The central treatment plant;
- The livestock farms where the waste is generated and disposed in the baseline;
- The site of the biogas combustion or energy generation facility;
- The manure storage tanks;
- The road itineraries and/or piping system between the manure collection points and the central treatment plant.

The relevant GHG sources included in or excluded from the project boundary are shown on the Table below.

Source		Gas	Included?	Justification/Explanation
Baseline	Direct emissions from the manure treatment processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted
		CH ₄	Yes	CH ₄ is the main source of the emissions in the baseline.
		N ₂ O	No	Minor N ₂ O emissions source in the baseline is not accounted for as per the applied methodology
		Other	-	-

Source		Gas	Included?	Justification/Explanation
	Emissions from electricity consumption / generation	CO ₂	Yes	Electricity is consumed from the grid in the baseline scenario.
		CH ₄	No	Excluded for simplification. This is conservative.
		N ₂ O	No	Excluded for simplification. This is conservative.
		Other	-	
	Emissions from thermal energy generation	CO ₂	No	Excluded for simplification. Although thermal energy out of waste heat obtained from gas engine's jacket is generated through the implementation of the proposed project activity, no emission is claimed for this, as noted in this PDD. Excluded for simplification. Although thermal energy out of waste heat obtained from gas engine's jacket will be generated through the implementation of the proposed project activity, no emission will be claimed for this, as noted in this PDD. It is simply because that, as per CDM approach, the heat used in the project scenario cannot be considered as baseline scenario since the thermal energy generated is kept within the project site without replacing any fossil-fuel-based source in the baseline scenarios. On the other hand, if the thermal energy generated at the project site was supplied to any other facility outside the project site, such as animal farm or a neighboring school or health center, the thermal energy replacing baseline fossil-fuel source of thermal used in such facilities -if such- would be regarded as baseline emissions. However, this is not the case in the project's reality. Hence, this exclusion shall be considered as appropriate.
		CH ₄	No	
		N ₂ O	No	
		Other	-	
Project	Emissions from thermal energy generation	CO ₂	No	Excluded for simplification. As noted above, the project also generates thermal power out of waste heat obtained from gas engine's jacket.. Although the proposed project activity does claim any emission reduction from the thermal energy generation, it does not consume thermal energy from any outside source either. The thermal energy generated within the boundaries of the proposed project activity is used for the project's auxiliary heat consumption. This exclusion shall be considered as appropriate.
		CH ₄	No	
		N ₂ O	No	
		Other	-	
	Emissions	CO ₂	Yes	These emissions are accounted for since the

Source	Gas	Included?	Justification/Explanation
from on-site electricity use			project may consume electricity from the national grid. In the case of consumption of electricity that is generated from collected biogas, emissions are not taken into consideration as per the applied Meth.
	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Other	-	-
	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
	CH ₄	Yes	The emissions from anaerobic digesters are accounted for.
	N ₂ O	No	Minor N ₂ O emissions source in the baseline in the baseline is not accounted for as per the applied Methodology
	Other	-	-
	CO ₂	Yes	It is an important emission source. Hence, they are accounted for.
	CH ₄	No	Excluded for simplification. This emission source is assumed to be very small.
	N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
	Other	-	-
Emissions from sludge composting	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for.
	CH ₄	No	The proposed project activity does not involve composting. Hence, this emission source is not accounted for.
	N ₂ O	No	The proposed project activity does not involve

Source		Gas	Included?	Justification/Explanation
				composting. Hence, this emission source is not accounted for.
		Other	-	-
	Emissions from manure storage tanks	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted for.
		CH ₄	No	Since manure is not stored more than 24 hours ⁴⁵ within the project's boundaries, these emissions are not accounted for.
		N ₂ O	No	Excluded for simplification. This emission source is assumed to be very small.
		Other	-	-

In addition, the diagram demonstrating the baseline and project scenario is given below. This diagram clearly shows the related emissions from baseline and project constituents.

⁴⁵ The AMS-III.D Small-scale Methodology: Methane recovery in animal manure management systems, version 21.0 states that project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if condition is satisfied: The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester.

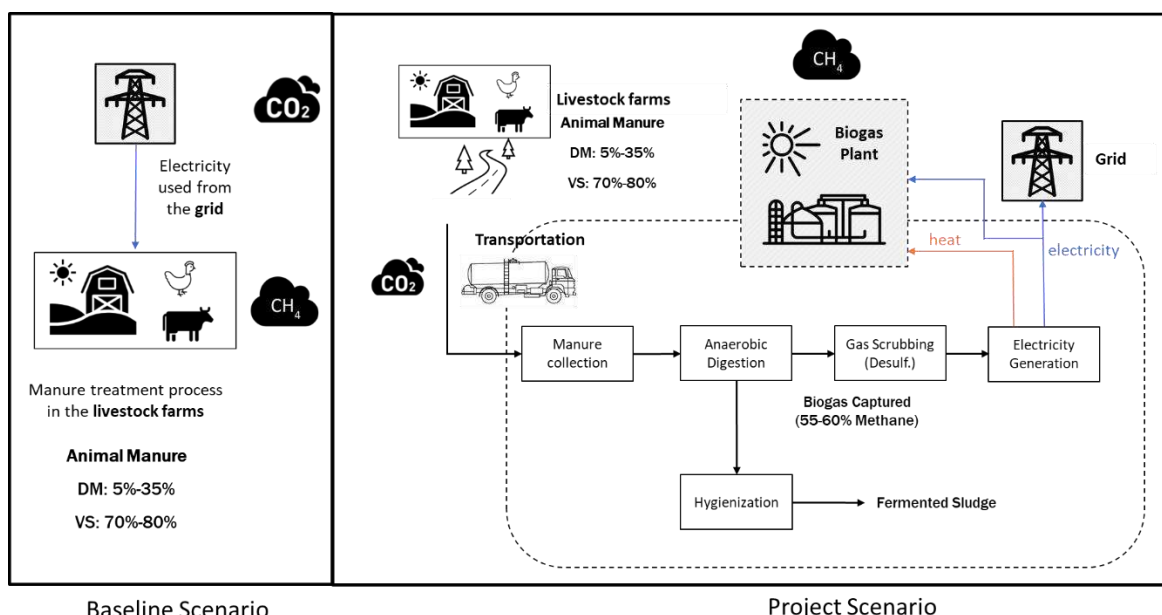


Figure 26. Baseline and Project Scenario

3.4 Baseline Scenario

The proposed project activity has two major GHG emission reduction components. The first one is the reduction of methane emissions resulting from improved animal manure management, and the second one is the use of biogas recovered from improved animal manure management for generating power and feeding the electricity generated partly to the national electricity grid network.

Primary Component of the Proposed Project Activity as per AMS-III.D., version 21.0

The primary component of the proposed project activity includes improved animal manure management and biogas recovery resulting in the decrease of the extent of methane emissions that emanate from the present mode of animal manure management at the farms whose manure is supplied to the project. And, hence the proposed project activity is considered as a replacement of the existing environmentally unfriendly animal manure management systems that results in anaerobic lagoon conditions and aims to achieve methane recovery and gainful use of recovered methane. Hence, the applicable baseline as per the approved methodology AMS-III.D., version 21.0, for the first component of the project activity is the following:

- The maximum methane-producing capacity of manure (Bo) varies by species and diet, and
- The organic material in livestock manure measured as Volatile Solids (VS) which consist of both biodegradable and non-biodegradable fractions.

The above parameters are used along with Global Warming Potential of methane, methane density, methane conversion factor, number and type of livestock, fraction of manure handled

and correction factors appropriately and the baseline emissions measured in t CO₂e is calculated in a conservative and transparent manner as per the approved methodology AMS-III.D., version 21.0.

With reference to AMS-III.D., version 21.0, for project activities involving methane recovery in animal manure management, the baseline scenario fits well the scenario defined in the paragraph 17 of the approved methodology:

“In the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere”.

The baseline methodology selected approach from paragraph 48 of the CDM modalities and procedures called “existing actual or historical emissions, as applicable” have been applied in the context of the project activity. The approach selected in the baseline methodology checks the additionality of the project activity and determines the baseline emission factor for selected baseline scenario.

The applicable baseline scenario is animal manure being left to decay anaerobically within the project boundary and methane being emitted directly to the atmosphere in the absence of the proposed project activity. The results of the baseline survey which is going to be presented in time of validation as documentary evidence, confirm that the common practice for livestock farm owners to treat manure generated at their farms is to store manure in uncovered anaerobic lagoons/ponds for several months till manure taking away. As such, the baseline scenario can be characterized as anaerobic lagoon.

Identification of Baseline Scenario

As per the paragraph 7 of the approved methodology AMS-III.D., version 21.0, the proposed project activity, as a Greenfield project, compared to the baseline scenario is only eligible if it complies with the related and relevant requirements in the "General guidelines for SSC CDM methodologies", version 23.0. The Section 4.11 of the Guidelines describes the “procedure to determine baseline scenario for Type II and Type III Greenfield projects” which is a 4-step process to be followed in order to determine whether the requirements are met.

Step 1: Identify the various alternatives to the project proponent that deliver comparable levels of service, including the proposed project activity undertaken without being registered as a CDM project activity.

The table 10.14 in Chapter 10, Volume 4 of 2019 Refinement of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories⁴⁶ provides a list of possible manure management systems for different livestock species, including dairy cattle, non-dairy cattle and poultry, three livestock species whose manure is treated in anaerobic digesters by the proposed project activity.

⁴⁶ https://www.ipcc-nggip.iges.or.jp/public/2019rf/pdf/4_Volume4/19R_V4_Ch10_Livestock.pdf (Accessed on 08/12/2023)

Manure management systems are listed as follows:

- Pasture/Range/Paddock
- Deep Bedding
- Composting
- Poultry manure with litter
- Poultry manure without litter
- Aerobic Treatment
- Liquid/Slurry
- Pit Storage under animal confinements
- Solid Storage
- Dry lot
- Burned for fuel
- Daily spread
- Anaerobic digestion

Uncovered anaerobic lagoon –Animal waste manure management prior to the project activity

The Regulation on Livestock Farms⁴⁷ implemented in 2006 in Turkey vaguely requests livestock farms to have a lagoon to collect the manure generated at the farms. While the regulation did not provide any technical specifications for such lagoons, it was stated that the lagoons' interior walls which stayed underground shall be of concrete foundation -in order to prevent the manure from leaking into soil and underground water resources. Thus, it is justifiable to state that there is currently no legal regulation for the livestock farms that defines an anaerobic lagoon design. Although it varies based on the scale of the farm, the overall depth of such lagoons at the farms around the project site is 2 meters. Since there is no legal regulation to mandate the livestock farm owners to capture and/or utilize methane generated at these lagoons, continuing to use anaerobic lagoons as manure management -without making any modification to their existing manure management systems, is viable and reasonable for animal farm owners.

Both liquid and solid manure generated at cattle and poultry farms are stored in the lagoons for 2-3 months till they are evacuated – usually to agricultural fields. This system complies with

⁴⁷ <https://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm> (Accessed on 08/12/2023)

environmental and legal regulations in Turkey and therefore applicable as a baseline and/or alternative scenario for the proposed project activity.

Pasture/Range/Paddock – Excluded due to regulatory requirements

The regulations in Turkey state that the animal manure should be collected in concrete made lagoons/ponds, this type of manure management system is not in compliance with the regulations.

Deep Bedding – Excluded due to regulatory requirements

This manure management system is not allowed due to the regulation that states manure shall be kept only on concrete made lagoons / ponds.

Composting – Excluded since it can only be applied as a secondary / continuous stage

Because the composting system can only be applied as a subsequent stage after manure storage and requires expertise, this technology is not applicable for the baseline scenario of this project.

Poultry manure with litter – Excluded due to regulatory requirements

This manure management system is not allowed due to the regulation that states manure shall be kept only on concrete made lagoons / ponds.

Poultry manure without litter – Excluded due to regulatory requirements

This manure management system is not allowed due to the regulation that states manure shall be kept only on concrete made lagoons / ponds.

Aerobic Treatment – Excluded due to high energy requirements and lack of sufficient land

Aerobic treatment is not a preferred manure management in Turkey since forced aeration requires high electricity consumption and natural aeration requires aerobic and facultative ponds and wetland systems which are limited in Turkey.

Liquid/Slurry – Excluded due to regulatory requirements

This manure management system is not allowed due to the regulation that states manure shall be kept only on concrete made lagoons / ponds.

Pit storage under animal confinements – Excluded due to regulatory requirements

This form of manure management was used to be applied in animal farms. However, in line with the legal regulations made on livestock farms, this form of manure management system is not allowed anymore. Those who had an installed system of this kind had given legal notice to reform their systems accordingly.

Solid storage – Excluded due to regulatory requirements

This manure management system is prohibited by the Regulation on Livestock Farms⁴⁸ which states that manure shall be stored on concrete made structures to prevent any harm on the environment and human health.

Dry lot – Excluded due to regulatory requirements

Because the manure management regulations in Turkey enforces the use of concrete storage areas, dry lot manure management system, which is a paved or unpaved open confinement area, is not a viable option for the livestock farm owners.

Burned for fuel – Excluded due to regulatory requirements

This manure management system is not in line with current regulations. Livestock farm owners are prohibited to use this manure management method because of the emission of hazardous gasses emitted from burning dung cakes.

Daily Spread – Excluded due to regulatory requirements

Current legal regulations in Turkey regarding livestock farms, this form of manure management system is not allowed since untreated animal manure contains high nitrogen level and harms the organic quality of soil as well as underground aquifers.

Anaerobic digestion (biogas) – The proposed project activity

This is the proposed project activity not being registered as a VERRA/VCS project activity. Due to barriers such as – technological barriers, lack of skilled labour to operate the facility and prevailing practice, as well as the investment barrier, this system is not applicable for the baseline scenario.

Step 3: Eliminate and rank the alternatives identified in Step 2 taking into account barrier tests specified in the “Guidelines on the demonstration of additionality of small-scale project activities”.

Uncovered anaerobic lagoon – Previous animal waste manure management

This has been the system that the livestock farm owners were utilizing. Since this system is commonly used, there are no barriers to analyze that would prevent the use of this system or its use continuation. As noted earlier, since there is no legal regulation to mandate the livestock farm owners to implement different anaerobic lagoon design and to capture and/or utilize

⁴⁸ <https://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm> (Accessed on 08/12/2023)

methane generated at these lagoons, this manure management system is the most economic, viable and reasonable for farm owners.

Anaerobic digestion (biogas) – The project activity undertaken without being registered as a CDM project activity – Eliminated due to technological and investment barriers

This alternative face both investment and technological barriers. Investment barriers relate to the increased levels of manure management complexity and significant capital outlay that are necessary for the installation and operation of anaerobic digester systems. Technological barriers relate to the complex nature of the anaerobic digestion technology that would require employment of a qualified engineering team. The sensitive nature of the process bears risks that if not operated carefully could lead to shut down of the process. Due the technological and investment barriers for local livestock farm owners, this alternative has to be implemented as a CDM Project Activity.

Step 4: If only one alternative remains that is not the proposed project activity undertaken without being registered as a CDM project activity and also corresponds to one of the baseline scenarios provided in the methodology, then the project activity is eligible under the methodology. If more than one alternative remains that correspond to a baseline scenario provided in the methodology, choose the alternative with the lowest emissions as the baseline.

Only one alternative remains that fulfils the requirements. The proposed project activity, as a Greenfield facility, shall, hence, be considered as eligible under methodology AMS-III.D., version 21.0, with the baseline scenario being uncovered anaerobic lagoons as manure management system.

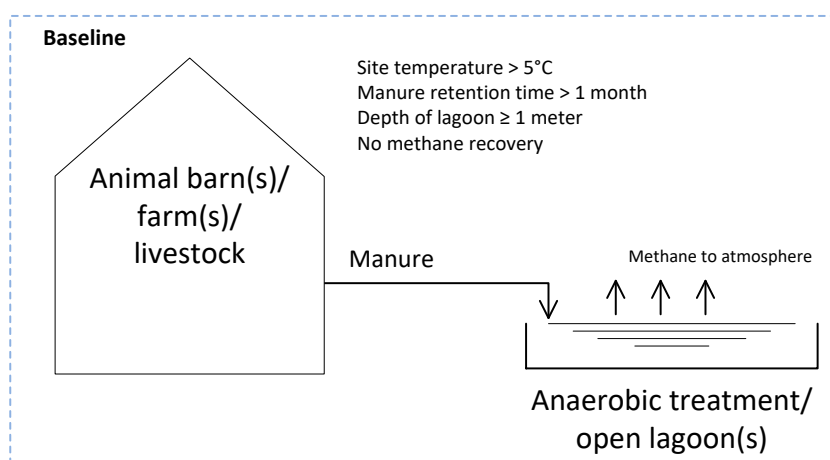


Figure 27. Baseline Scenario

As per AMS III.D, baseline emissions of the uncovered lagoon are calculated using option (b), Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its

specific volatile solids (SVS) content. See section 4 for detailed information regarding calculation method, formula and parameters.

Second Component of the Proposed Project Activity as per AMS-I.D., version 18.0

The second component of the proposed project activity includes feeding electricity so generated from the renewable power plant to the national electricity grid network in Turkey. And, hence the renewable power plant can be considered as the installation of new grid-connected renewable power plant that feed the electricity for a grid system, which is fed by both fossil fuel fired generating plants (using fossil fuels such as coal, natural gas, diesel etc.) and non-fossil fuel based generating plants (such as hydro, nuclear, solar, biomass and wind).

The paragraph 19 of AMS-I.D., version 18.0, defines the baseline scenario for greenfield power plant, like in the case of the project activity, as “the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid”.

The baseline methodology selected approach from paragraph 48 of the CDM modalities and procedures called “existing actual or historical emissions, as applicable” have been applied in the context of the project activity. The approach selected in the baseline methodology checks the additionality of the project activity and determines the baseline emission factor for the selected baseline scenario.

The project will generate electricity using biogas recovered from animal manure. Hence the biogas-based power generated from the project site will be replacing the electricity generated from the fossil fuel power stations feeding into the national grid. Since the source of the power has reduced GHG emissions, the power generated will save the anthropogenic GHG emissions that would have been generated by the fossil fuel based thermal power stations comprising coal, diesel, furnace oil and gas. The estimation of GHG reductions by the second component of the project activity is limited to CO₂ only.

The methodology lays down certain steps by which the baseline is determined. The baseline methodology identifies the project as being additional and not the baseline scenario. The proposed project activity will impact the operating margin, build margin and thus the combined margin. The methodology is designed to have the grid combined margin as the baseline scenario. For grid connected electricity generation projects it is important to ascertain whether the proposed project activity has some impact on the national grid's electricity generating pattern. It has been established in the CDM modalities and procedures that the use of the combined margin which takes into account the operating and build margins can be used to determine the effect of the power project to the grid.

As per AMS I.D., version 18.0, the baseline emission of the power generation is the MWh produced by the renewable generating unit multiplied by an emission coefficient (measured in kg CO₂e/MWh) calculated in a transparent and conservative manner. The emission coefficient

is calculated according to option (a) selected from the methodology AMS-I.D., version 18.0, as: (a) A combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the 'Tool to calculate the emission factor for an electricity system'. The detailed calculation formulae and data source used are also provided in section 4.

3.5 Additionality

Section 3.13 of VCS Standard, v4.4, clearly mandates that "Additionality shall be demonstrated and assessed in accordance with the requirements set out in the methodology applied to the project".

The paragraph 15 of the approved methodology of AMS-III.D., version 21.0, applied to the proposed project activity, states that "Project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the 'Guidelines on the demonstration of additionality of small-scale project activities'".

As noted above, there is no legal regulation in Turkey that requires the collection and destruction of methane from livestock manure. Thus, it is not required to apply the 'Guidelines on the demonstration of additionality of small-scale project activities'. Accordingly, Type III component of the proposed project activity, as methane recovery from animal manure, shall be considered as deemed additional. Further, this demonstrates the regulatory surplus of the project activity which is defined as "Project activities are not mandated by any law, statute, or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute, or other regulatory framework." in the VCS Program Definitions v4.449.

Moreover, paragraph 16 of AMS-III.D., version 21.0. clarifies that "This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component".

Within the boundaries of the Type III component of the proposed project activity, the project captures and utilizes biogas from animal manure in anaerobic digesters to generate electricity with biogas engines/units with the total installed capacity 1.5 MW, lower than 5 MW. Hence, as per the paragraph 16 of AMS-III.D., version 21.0, it is not required to apply the 'Guidelines on the demonstration of additionality of small-scale project activities' to the Type I component of the proposed project activity which is grid-connected renewable electricity generation.

49 <https://verra.org/wp-content/uploads/2023/08/VCS-Program-Definitions-v4.4-updated-4-Oct-2023.pdf> (Accessed on 08/12/2023)

Accordingly, Type I component of the proposed project activity, as grid-connected renewable electricity generation, shall also be considered as deemed additional.

Therefore, the proposed project activity, as per the approved methodology of AMS-III.D., version 21.0, is automatically defined as additional, without further documents of barriers as per “Guidelines on the demonstration of additionality of small- scale project activities”.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Calculation of Baseline Emissions as per AMS-III.D.

As per paragraph 17 of AMS-III.D., version 21.0., the baseline scenario is the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere. To calculate the baseline emissions, the project proponents propose to utilize the Option 17 (b) presented in the AMS-III.D., version 21.0., Paragraph 17. This is because the project proponents are aware of the existing animal manure management systems i.e., anaerobic lagoon and the IPCC guidelines are utilized for determining the manure characteristics. The available laboratory facility of the project enables the direct measurement of the specific volatile solids (SVS) parameter, that is measured periodically and aggregated to find yearly for each of the farm. In addition, parameter ($Q_{\text{manure},j,LT,y}$), being the quantity of manure treated for each farm and animal type is measured through weighbridge, continuously, and aggregated to find monthly and yearly. The direct measurement of these parameters allowed the PP to use Option Para 17(b) as follows: As per AMS III.D., version 21.0.

Baseline emissions

Calculation of Baseline Emissions as per AMS-III.D.

In accordance with the methodology AMS III.D., version 2, baseline emissions were calculated based on the direct measurement of quantity of manure and its specific volatile solids content, as follows:

Equation (1)

$$BE_y = BE_{CH_4} = GWP_{CH_4} \times D_{CH_4} \times UF_b \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times Q_{manure,j,LT,y} \times SVS_{j,LT,y})$$

Where:

BE_y	= Total baseline emissions in year y, in tCO ₂ e/year
GWP_{CH_4}	= Global Warming Potential of CH ₄ applicable to the crediting period (28 tCO ₂ e/tCH ₄)
D_{CH_4}	= CH ₄ density (0.00067 t/m ³ at room temperature (20°C) and 1 atm pressure)
UF_b	= Model correction factor to account for model uncertainties (0.94)
LT	= Index for all types of livestock
j	= Index for animal manure management system
MCF_j	= Annual methane conversion factor for the baseline animal manure management system j
$B_{0,LT}$	= Maximum methane producing potential of the volatile solid generated for animal type LT (m ³ CH ₄ /kg-dm)
$Q_{manure,j,LT,y}$	= Quantity of manure treated from livestock type LT and animal manure management system j (tonnes/year, dry basis)
$SVS_{j,LT,y}$	= Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y (tonnes/tonnes, dry basis)

Estimation of $B_{0,LT}$ and MCF_i :

According to the applied methodology, Maximum Methane Production Potential ($B_{0,LT}$) can be determined in two different ways including the use of default values from the IPCC Guidelines and direct measurement. The project participants choose to determine in the way that default values are used, and they are obtained from 2019 IPCC Refinement, as updated in July 2023, Volume 4, Chapter 10, Table 10.16A (Updated), (page 10.67, Other Regions, High Productivity Systems -specific to the country where the proposed project activity is implemented). The direct measurement method is eliminated due to a lack of data.

The values for Methane conversion factors (MCF_i) are obtained from 2019 IPCC Refinement, as updated in July 2023, Volume 4, Chapter 10, Table 10.17 (Updated), (page 10.68), depending on the annual average temperature where the anaerobic manure treatment facility in the baseline is located. Future revisions to the IPCC Guidelines for National Greenhouse Gas Inventories are considered. Note that a conservativeness factor is also applied by multiplying MCF values (estimated as per above) with a value of 0.94, to account for the 20% uncertainty in the MCF values as reported by IPCC 2006.

The other fixed parameters are given in the table below:

Parameter	Value	Unit
GWP _{CH4}	28	Dimensionless
D _{CH4}	0.00067	t/m ³
MCF _j	0.76	Fraction
B _{0,Other Cattle}	0.18	m ³ CH ₄ / kg-dm
UF _b	0.94	Fraction

Accordingly,

BE_{CH4} is calculated as **30,188 tCO₂/yr.**

Calculation of Baseline Emissions as per AMS-I.D.

As per paragraph 22 of AMS-I.D, version 18., baseline emissions include only; CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

Equation (2)

$$BE_y = BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y}$$

Where:

BE_y = Total baseline emissions in year y, (tCO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of VCS project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh)

Since the proposed project activity is the installation of a greenfield power plant, as per the paragraph 26 of AMS-I.D, version 18., $EG_{PJ,y}$ is calculated as follows:

Equation (3)

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

$EG_{PJ, facility, y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y

As per paragraph 23 of AMS-I.D, version 18., the emission factor shall be calculated in a transparent and conservative manner. The project participants choose to calculate a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system”, version 07.0.

According to the Tool, the “combined margin” emission factor is calculated through the following six steps as per the “Tool to calculate the emission factor for an electricity system”, version 07.0:

- Step (1): Identify the relevant electricity system;
- Step (2): Choose whether to include off-grid power plant in the project electricity system (optional);
- Step (3): Select a method to determine the operating margin (OM);
- Step (4): Calculate the operating margin emission factor according to the selected method;
- Step (5): Calculate the build margin (BM) emission factor;
- Step (6): Calculate the combined margin (CM) emission factor.

Step 1: Identify the relevant electricity system

According to the applicability of “Tool to calculate the emission factor for an electricity system”, version 07.0, the relevant project electricity system should be identified by the project participants. The chosen electric power system is the Turkish National Grid, which is physically connected through transmission and distribution lines to the project activity. National grid is dominated by fossil fuels such as coal, diesel and natural gas.

Step 2: Choose whether to include off-grid power plants in the project electricity system (optional)

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, project participants may choose between the following two options to calculate the operating margin and build margin emission factor:

- Option I: Only grid power plants are included in the calculation.
- Option II: Both grid power plants and off-grid power plants are included in the calculation.

Step 3: Select a method to determine the operating margin (OM)

- Simple OM; or
- Simple adjusted OM; or
- Dispatch data analysis OM; or
- Average OM.

Step 4: Calculate the operating margin emission factor according to the selected method

The simple OM may be calculated by one of the following two options:

Option B: Based on the total net electricity generation of all power plants serving and the fuel types and total fuel consumption of the project electricity system.

$$EF_{grid,OM,y} = 0.7424 \text{ tCO}_2/\text{MWh}$$

Step 5: Calculate the build margin (BM) emission factor

⁵⁰<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf> (Accessed on 13/12/2022)

Option 1: for the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of PDD submission to the DoE for validation.

Option 2: for the first crediting period, the build margin emission factor shall be updated annually, ex post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available.

For the proposed project activity, the project participants choose Option 1 in terms of availability of vintage of data.

Based on 2020 data, Ministry of Energy and Natural resources of Turkey has calculated build margin and published the data:

$$EF_{grid,BM,y} = 0.3680 \text{ tCO}_2/\text{MWh}$$

Step 6: Calculate the combined margin emissions factor

According to the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the calculation of the combined margin (CM) emission factor ($EF_{grid,CM,y}$) is based on one of the following methods:

The combined margin emissions factor is calculated as follows:

Equation (4)

$$EF_{grid,CM,y} = EF_{grid,OM,y} \times w_{OM} + EF_{grid,BM,y} \times w_{BM}$$

Where:

$EF_{grid,BM,y}$ = Build margin CO₂ emission factor in year y (tCO₂/MWh)

$EF_{grid,OM,y}$ = Operating margin CO₂ emission factor in year y (tCO₂/MWh)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor in year y (tCO₂/MWh)

w_{OM} = Weighted of operating margin emissions factor (per cent)

w_{BM} = Weighted of build margin emissions factor (per cent)

As per the applicable “Tool to calculate the emission factor for an electricity system”, version 07.0, the default weights for the operating margin and build margin emission factors for the proposed project activity is defined as:

$$w_{OM} = 0.50$$

$$w_{BM} = 0.50$$

Based on 2020 data, Ministry of Energy and Natural resources of Turkey has calculated combined margin and published the data:

$$EF_{grid,CM,y} = 0.5552^{51} \text{ tCO}_2/\text{MWh}$$

Accordingly;

$$BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y}$$

$$BE_{EC,y} = 6,658 \times 0.5552$$

$$BE_{EC,y} = 3,696 \text{ tCO}_2\text{e/yr}$$

Accordingly, summing up the baseline emissions for the Type III and Type I components;

$$BE_y = BE_{CH_4,y} + BE_{EC,y}$$

$$BE_y = 30,188 + 3,966$$

$$BE_y = 33,884 \text{ tCO}_2/\text{yr}$$

4.2 Project Emissions

Calculation of Project Emissions as per AMS-III.D.

In accordance with AMS-III.D., version 21.0., Project activity emissions consist of:

Equation (5)

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Where;

PE_y = Project emissions in year y (t CO₂e)

$PE_{PL,y}$ = Emissions due to physical leakage of biogas in year y (t CO₂e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the biogas stream in the year y (t CO₂e)

$PE_{power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)

⁵¹ ⁵¹ It is value that obtained in time of validation and obtained from the most recent official national grid emission factor data published by the Ministry of the Energy and Natural Resources in Turkey. This value might be subject to a change in time of verification in accordance with the revisions of the Ministry of the Energy and Natural Resources in Turkey.

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUES EmisyonFktr/Belgeler/Bform2020.pdf> (Accessed on 13.12.2022)

$PE_{transp,y}$ = Emissions from incremental transportation in the year y (t CO₂e), as per relevant paragraph in AMS-III.AO

$PE_{storage,y}$ = Emissions from the storage of manure (t CO₂e)

As per the paragraph 21(b) of AMS-III.D., version 21.0., project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect, and transport the biogas to the point of flaring or gainful use are determined as, in accordance with the paragraph 23 of the methodological tool “Project and leakage emissions from anaerobic digesters, version 02.0. The PP has chosen to apply the Option 2 in the applied Tool to calculate the quantity of methane produced in the digester.

Equation (6)

$$Q_{CH_4} = Q_{biogas,y} \times f_{CH_4,default} \times \rho_{CH_4}$$

Equation (7)

$$PE_{PL,y} = PE_{CH_4,y} = Q_{CH_4} \times EF_{CH_4,default} \times GWP_{CH_4}$$

Where;

$PE_{PL,y}$ = Emissions due to physical leakage of biogas in year y (t CO₂e)

$PE_{CH_4,y}$ = Project emissions of methane from the anaerobic digester in year y (t CO₂e)

$Q_{CH_4,y}$ = Quantity of methane produced in the anaerobic digester in year y (t CH₄)

$EF_{CH_4,default}$ = Default emission factor for the fraction of CH₄ produced that leaks from the anaerobic digester (fraction)

GWP_{CH_4} = Global warming potential of CH₄ (t CO₂/t CH₄)

The parameter $EF_{CH_4,default}$ is determined according to the applied tool “Project and leakage emissions from anaerobic digesters, version 02.0. The default factor for Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction has been taken into account.

Accordingly;

$$EF_{CH_4,default} = 0.028$$

$$Q_{CH_4,y} = 3,197,940 \text{ N m}^3 \text{ biogas} \times 0.00067 \text{ t/m}^3 \times 0.6 \text{ Nm}^3 \text{ CH}_4 / \text{Nm}^3 \text{ biogas}$$

$$Q_{CH_4,y} = 1,286 \text{ tCH}_4$$

$$GWP_{CH_4} = 28$$

Consequently;

$$PE_{PL,y} = Q_{CH_4} \times EF_{CH_4,default} \times GWP_{CH_4}$$

$$PE_{PL,y} = 1,286 \times 0.028 \times 28$$

$$PE_{PL,y} = 1,008 \text{ tCO}_2\text{e}$$

As per the paragraph 23 of AMS-III.D., version 21.0., project emissions from electricity and fossil fuel consumption are determined by following the methodological tool “Project and leakage emissions from anaerobic digesters”, version 02.0., where $PE_{power,y}$ is the sum of $PE_{EC,y}$ and $PE_{FC,y}$ in the tool.

The related methodological tool also states that “If the electricity consumed is generated on-site using biomass residues, wind, hydro or geothermal power, then $PE_{EC,y} = 0$. Otherwise, the project can calculate $PE_{EC,y}$ using the monitored data or using a default value for electricity consumption from the grid. In the case where monitored data is used, $PE_{EC,y}$ shall be calculated using the “Tool to calculate baseline, project and/or leakage emissions from electricity consumption”, where the project emission source j referred to in the tool is the total electricity consumption associated with the anaerobic digestion facility.

Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 03.0. In accordance with the Tool, in the generic approach, project emissions from consumption of electricity are calculated based on the quantity of electricity consumed, an emission factor for electricity generation and a factor to account for transmission losses, as follows:

Equation (8)

$$PE_{power} = PE_{EC,y} = \sum_j (EC_{PJ,j,y} \times EF_{EF,j,y} \times (1 + TDL_{j,y}))$$

$EC_{PJ,j,y}$ = Quantity of electricity consumed by the project electricity consumption source j in year y (MWh/yr)

$EF_{EF,j,y}$ = Emission factor for electricity generation for source j in year y (t CO₂/MWh)

$TDL_{j,y}$ = Average technical transmission and distribution losses for providing electricity to source j in year y

However, because the project activity supplies the electricity consumption associated with the anaerobic digestion facility from only the on-site electricity generation, no electricity consumption from national grid is involved. Accordingly;

$$PE_{EC,y} = 0$$

In case the electricity consumes from the national grid due to any circumstance, such as emergency, the relevant calculations will be included in the upcoming monitoring period.

As per the paragraph 22 of the methodological tool “Project and leakage emissions from anaerobic digesters”, version 02.0., which determines project emissions from fossil fuel consumption ($PE_{FC,y}$), states that where the anaerobic digester facility uses fossil fuels, project participants shall calculate $PE_{FC,y}$ using the “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”. However, since in the proposed project activity the anaerobic digester facility does not use fossil fuels, $PE_{FC,y}$ is accounted as zero. Thus, PE_{power} is also accounted as zero.

Project emissions from flaring shall be calculated using the following equation:

Equation (9)

$$PE_{flare,y} = GWP_{CH_4} \times \sum_{m=1}^{525600} F_{CH_4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

Where:

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO₂e)

GWP_{CH_4} = Global warming potential of methane valid for the commitment period (tCO₂e/tCH₄)

$F_{CH_4,RG,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,m}$ = Flare efficiency in the minute m

In accordance with the paragraph 22 of AMS-III.D., version 21.0., in the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool “Project emissions from flaring”, version 04.0. If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.

Within the project activity captured methane is used for electricity generation through the gas engine. Hence, the PP considers a methane destruction efficiency of 100% in the gas engine. In line with the para 22 of the applied methodology, output of the gas engine (generated electricity) is monitored.

In the proposed project activity, an enclosed flare is installed. The flare is installed for the case of emergency and malfunction in the plant and in order to ensure no biogas is released under exigencies. Within the scope of the monitoring plan, the quantity of captured methane fed to

the flare is measured and recorded. This emergency flare system is not used under normal operation. In addition, in case this emergency flare operated, the emissions reductions will be excluded for conservativeness. Thus $PE_{flare,y}$ is also excluded. Moreover, flare is equipped with sensors that detect and response in the presence of a flame. All these monitoring activities has been recorded both in operational logbook and as electronic data by the staff. PP has as well checked such recordings and confirmed that no flaring activity was observed in time of validation. Therefore, there is no project emissions from flaring.

Nevertheless, the monitoring plan of the facility shall measure the quantity of captured methane fed to the flare. If such measurement is positive, the emissions from flaring the biogas shall be determined following the procedure described in the methodological tool project emissions from flaring, version 04.0.

As per the paragraph 24 of AMS-III.D., version 21.0., project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

- (a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;
- (b) The dry matter content of the manure when removed from the animal barns is less than 20%.

Since the storage time of the manure after removal from the animal barns, including transportation, does not exceed 24 hours before being fed into the anaerobic digester in the proposed project activity, $PE_{storage,y}$ shall be accounted as zero.

In accordance with AMS-III.D., version 21.0., $PE_{trans,y}$ is calculated as per the paragraph 14 of AMS-III.AO, version 01.0., which states project emissions due to incremental transport distances ($PE_{trans,y}$) are calculated based on the incremental distances between:

Equation (10)

$$PE_{trans,y} = (Q_y/CT_y) \times DAF_w \times EF_{CO2,transport}$$

Where:

Q_y = Quantity of raw waste/manure treated and/or wastewater co-digested in the year y (tonnes)

CT_y = Average truck capacity for transportation (tonnes/truck)

DAF_w = Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)

$EF_{CO2,transport}$ = CO₂ emission factor from fuel use due to transportation (kgCO₂/km, IPCC default values or local values may be used)

CO₂ emission factor from fuel use due to transportation is determined using the Methodological Tool “Project and leakage emissions from road transportation of freight” Version 01.0.0. As per the applied tool, the vehicle class is defined as light vehicle if the gross vehicle mass is less or equal to 26 tonnes. Since the gross vehicle mass for the vehicles is 21 tonnes, the emission factor for light vehicles is taken. Accordingly;

$$Q_y = 112,430 \text{ tonnes}$$

$$CT_y = 21 \text{ tonnes/truck}$$

$$DAF_w = 14.3 \text{ km/truck}$$

$$EF_{CO_2,transport} = 0.245 \text{ kgCO}_2/\text{km}$$

$$PE_{trans,y} = (Q_y/CT_y) \times DAF_w \times EF_{CO_2,transport}$$

$$PE_{trans,y} = (112,430/21) \times 14.3 \times 0.245$$

$$PE_{trans,y} = 19 \text{ tCO}_2/\text{y}$$

Accordingly, the project emissions for the Type III component is as follows:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

$$PE_y = 1,008 + 0 + 0 + 19 + 0$$

$$PE_y = 1,027 \text{ tCO}_2\text{e/yr}$$

4.3 Leakage

As per the applied methodology, leakage is determined by following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”, version 02.0⁵². The tool accounts for leakage emissions associated with storage and composting of digestate. Within the scope of the project activity, digestate obtained from biogas production is distributed free of charge to the neighboring cropland and farm owners without any storage under anaerobic conditions or composting process. In other words, in Section 2.2 of the methodological tool “Project and leakage emissions from anaerobic digesters”, it is stated as follows:

“(a) CH₄ and N₂O emission from composting of digestate;

(b) CH₄ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond”.

Considering the methodological tool, the project activity does not involve composting of the digestate or the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic

⁵² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-14-v2.pdf> (Accessed on 08/12/2023)

storage, such as in a stabilization pond. Therefore, leakage emissions associated with the anaerobic digesters are not taken into account for the project activity.

4.4 Net GHG Emission Reductions and Removals

Emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emission in year y (tCO₂e/yr)

PE_y = Project emission in year y (tCO₂e/yr)

For the Type III components:

In accordance with the methodology AMS III.D., version 2, baseline emissions were calculated based on the direct measurement of quantity of manure and its specific volatile solids content (para 17b), as follows:

$$BE_y = BE_{CH_4} = GWP_{CH_4} \times D_{CH_4} \times UF_b \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times Q_{manure,jLT,y} \times SVS_{j,LT,y})$$

Farm's Name	Type of Livestock (LT)	BE _{CH₄,y} (tCO ₂ e/yr/farm)
Ali Bey Çiftliği (Necmettin Erden)	Other Cattle	2,300
Erol Yılmaz	Other Cattle	2,023
Bey Besi (Veysel Karabey)	Other Cattle	2,216
Şenol Yılmaz Besi Çiftliği	Other Cattle	535
Saygın Bey (Veysel Karabey)	Other Cattle	2,947
Aytekin Besi	Other Cattle	1,130
Has Besi	Other Cattle	9,200
İzzet Alp	Other Cattle	5,539
yılmazlar besi (kerim yılmaz)	Other Cattle	4,299
TOTAL		30,188

Consequently;

BE_{CH4,y} is calculated as **30,188 tCO_{2e}/yr**

In accordance with AMS-III.D., version 21.0., Project activity emissions consist of:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

$$PE_y = 1,008 + 0 + 0 + 19 + 0$$

$$PE_y = 1,027 \text{ tCO}_2\text{e/yr}$$

Consequently;

PE_y is calculated as **1,027 tCO_{2e}/yr**

Consequently, net ER considering type III components is calculated as follows:

$$ER_{y, \text{ Type III}} = BE_{CH4,y} - PE_y$$

$$ER_{y, \text{ Type III}} = 30,188 - 1,027$$

$$ER_{y, \text{ Type III}} = 29,161 \text{ tCO}_2\text{e/yr}$$

For the Type I components:

$$ER_{y, \text{ Type I}} = BE_{EC,y} - PE_y$$

$$ER_{y, \text{ Type I}} = 3,696 - 0$$

$$ER_{y, \text{ Type I}} = 3,696 \text{ tCO}_2\text{e/yr}$$

Considering both the Type III and Type I Components:

$$ER_y = ER_{y, \text{ Type III}} + ER_{y, \text{ Type I}}$$

$$ER_y = 29,161 + 3,696$$

$$ER_y = 32,857 \text{ tCO}_2\text{e/yr}$$

The estimated emission reduction was calculated over a 365-day period.

The estimated net emission reductions during the first crediting period are as follows:

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
31/12/2020	93	3	0	90
01/01/2021-31/12/2021	33,884	1,027	0	32,857

01/01/2022-31/12/2022	33,884	1,027	0	32,857
01/01/2023-31/12/2023	33,884	1,027	0	32,857
01/01/2024-31/12/2024	33,884	1,027	0	32,857
01/01/2025-31/12/2025	33,884	1,027	0	32,857
01/01/2026-31/12/2026	33,884	1,027	0	32,857
01/01/2027-30/12/2027	33,791	1,024	0	32,767
Total	237,188	7,189	0	229,999

Furthermore, in accordance with the applied methodology, the emission reductions achieved by the project activity will be determined ex post through direct measurement of the amount of methane fueled, flared or gainfully used. Ex-post emission reductions achieved in any year are determined as the lowest value of the following:

$$ER_{y,ex-post} = \min[(BE_{y,ex-post} - PE_{y,ex-post}), (MD_y - PE_{power,y,ex-post})]$$

Where;

$ER_{y,ex-post}$ = Emission reductions achieved by the project activity based on monitored values for year y (t CO₂e)

$BE_{y,ex-post}$ = Baseline emissions calculated using equation 1

$PE_{y,ex-post}$ = Project emissions calculated using equation 6 using ex post monitored

Values

MD_y = Methane captured and destroyed or used gainfully by the project activity in year y (t CO₂e)

$PE_{power,y,ex-post}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y (t CO₂e)

As per para 30 of the applied Methodology, if project activities utilize the recovered methane for power generation, MD_y will be calculated based on the amount of monitored electricity generation as follows:

$$MD_y = \frac{EG_y}{NCV_{CH_4} * EE_y} * D_{CH_4} * GWP_{CH_4}$$

Where;

EG_y = Total electricity generated from the recovered biogas in year y (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/Nm³) (use default value: 35.9 MJ/Nm³)

EE_y = Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:

- Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation;
- Default efficiency of 40 %.

As this project involves gainful use of captured methane through electricity generation, the project proponents will be using the above method for the calculation of ex-post emission reductions in the upcoming monitoring periods. The amount of generated electricity is continuously monitored through electricity meters and recorded on a monthly basis.

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	ρ_{CH_4}
Data unit	t/m ³
Description	Density of methane at normal (at room temperature 20 °C and 1 atm pressure) conditions
Source of data	Approved Large-Scale Methodology of AM0073: GHG emission reductions through multi-site manure collection and treatment in a central plant, Version 1.0 ⁵³

⁵³ https://cdm.unfccc.int/UserManagement/FileStorage/CDMWF_AM_8CTM5MFZLTDO7SZK1A6D7AK3YPIG7S
(Accessed on 08/12/2023)

Value monitored	0.00067
Justification of choice of data or description of measurement methods and procedures applied	0.00067 t/m ³ at room temperature 20°C and 1 atm pressure
Purpose of Data	Used in • Calculation of baseline emissions • Calculation of project emissions
Comments	6.7x10 ⁻⁴ t/m ³ equals to 0.00067 t/m ³

Data / Parameter	$f_{CH_4, default}$
Data unit	Nm ³ CH ₄ / Nm ³ biogas
Description	Default value for the fraction of methane in the biogas
Source of data	Tool14: Project and leakage emissions from anaerobic digesters, version 02.0 (page 11/16)
Value applied:	0.6
Justification of choice of data or description of measurement methods and procedures applied	Default value from applied tool is used as per the applied methodology and as per Version 4. 5 ⁵⁴ of the VCS Standard
Purpose of Data	Used in • Calculation of project emissions
Comments	In line with the tool, this value for Option 2 of the step “Determination of the quantity of methane produced in the digester” is utilized for calculations.

⁵⁴ <https://verra.org/wp-content/uploads/2023/08/VCS-Standard-v4.5.pdf> (Release Date 29/08/2023) (Accessed on 08/12/2023)

Data / Parameter	GWP_{CH_4}
Data unit	Dimensionless
Description	Global Warming Potential for CH ₄
Source of data	IPCC Fifth Assessment Report, Chapter 8, Table 8.A.1 ⁵⁵ (page 731)
Value applied:	28
Justification of choice of data or description of measurement methods and procedures applied	Default value from IPCC is used as per the applied methodology and as per Version 4. 5 ⁵⁶ of the VCS Standard
Purpose of Data	Used in • Calculation of baseline emissions • Calculation of project emissions
Comments	In Version 4.1 of the VCS Standard, an update to VCS rules requires use of global warming potential values from the <u>IPCC Fifth Assessment Report</u> (updating from the Fourth Assessment Report) for calculating greenhouse gas (GHG) emission reductions. This update is consistent with the Paris Agreement and other carbon crediting programs ⁵⁷ .

Data / Parameter	UF_b
Data unit	Fraction
Description	Model correction factor

⁵⁵ https://www.ipcc.ch/site/assets/uploads/2018/02/WG1AR5_all_final.pdf (Accessed on 08/12/2023)

⁵⁶ <https://verra.org/wp-content/uploads/2023/08/VCS-Standard-v4.5.pdf> (Release Date 29/08/2023) (Accessed on 08/12/2023)

⁵⁷ <https://verra.org/vcs-standard-v4-1-will-scale-up-finance-for-climate-mitigation/> (Accessed on 08/12/2023)

Source of data	AMS-III.D. Version 21.0, paragraph 18 ⁵⁸ (page 7/30)
Value monitored	0.94
Justification of choice of data or description of measurement methods and procedures applied	Model correction factor to account for model uncertainties
Purpose of Data	Used in Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{CH_4, default}$
Data unit	Fraction (t CH ₄ leaked / t CH ₄ produced)
Description	Default emission factor for the fraction of CH ₄ produced that leaks from the anaerobic digester
Source of data	Methodological Tool 14: Project and leakage emissions from anaerobic digesters, Version 02.0 (page 12/16)
Description of measurement methods and procedures to be applied	Digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction.
Frequency of monitoring/recording	-
Value monitored	0.028
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Used in

⁵⁸ <https://cdm.unfccc.int/UserManagement/FileStorage/1AWXEKHVTF423LCN56Z9GIMQOS8JR> (Accessed on 08/12/2023)

	Calculation of project emissions
Calculation method	-
Comments	The default value is determined in accordance with the type of digesters used in the project activity, defined by CDM Tool 14, version 02.0.

Data / Parameter	EF _{CO₂,transport}
Data unit	kgCO ₂ /km
Description	CO ₂ emission factor from fuel use due to transportation
Source of data	Methodological tool: Project and leakage emissions from transportation of freight, Version 01.0 (page 4)
Description of measurement methods and procedures to be applied	Default value for light vehicles is selected. Selection is conservative compared to heavy vehicles and in agreement with the type of vehicles to be used for the project transportation activities.
Frequency of monitoring/recording	-
Value monitored	0.245
Monitoring Equipment	-
QA/QC procedures to be applied	-
Purpose of Data	Used in Calculation of project emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	Q _{manure,LT,y}																			
Data unit	Tonnes-dm/year																			
Description	Quantity of manure treated from livestock type LT at animal manure management system j																			
Source of data	On site measurement, operational and lab logbooks (weighbridge and lab logbook records)																			
Description of measurement methods and procedures to be applied	Each manure load transported to the site is measured with weight balance / or truck load capacity. (In case of weighbridge failure, rated capacity of truck load is considered for the calculation.) In addition, since the applied Meth requires quantity of manure to be measured in dry basis, the weighbridge data is multiplied with dry matter parameter measured for each farm.																			
Frequency of monitoring/recording	Annually, based on daily measurement and monthly aggregation.																			
Value monitored	<table><tr><th>Farm Name</th><th>Q_{manure,LT,y} (tonnes-dm/yr)</th></tr><tr><td>Ali Bey Çiftliği (Necmettin Erden)</td><td>1,192</td></tr><tr><td>Erol Yılmaz</td><td>1,022</td></tr><tr><td>Bey Besi (Veysel Karabey)</td><td>1,225</td></tr><tr><td>Şenol Yılmaz Besi Çiftliği</td><td>292</td></tr><tr><td>Saygın Bey (Veysel Karabey)</td><td>1,745</td></tr><tr><td>Aytekin Besi</td><td>689</td></tr><tr><td>has besi</td><td>5,297</td></tr><tr><td>izzet alp</td><td>3,021</td></tr></table>		Farm Name	Q _{manure,LT,y} (tonnes-dm/yr)	Ali Bey Çiftliği (Necmettin Erden)	1,192	Erol Yılmaz	1,022	Bey Besi (Veysel Karabey)	1,225	Şenol Yılmaz Besi Çiftliği	292	Saygın Bey (Veysel Karabey)	1,745	Aytekin Besi	689	has besi	5,297	izzet alp	3,021
Farm Name	Q _{manure,LT,y} (tonnes-dm/yr)																			
Ali Bey Çiftliği (Necmettin Erden)	1,192																			
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	<table> <tr> <td>yılmazlar besi (kerim yılmaz)</td><td>2,285</td></tr> <tr> <td>TOTAL</td><td>16,767</td></tr> </table>	yılmazlar besi (kerim yılmaz)	2,285	TOTAL	16,767								
yılmazlar besi (kerim yılmaz)	2,285												
TOTAL	16,767												
Monitoring equipment	Weighbridge scales at the entrance and/or truck load capacity and trip aggregation. Also, lab logbook is used to obtain DM parameter for each farm. <table> <tr> <td>Name:</td><td>Weighbridge</td></tr> <tr> <td>Manufacturer:</td><td>ESİT</td></tr> <tr> <td>Type:</td><td>Non-Automatic weighbridge with a capacity of 60,000-80,000 kg</td></tr> <tr> <td>Accuracy Class:</td><td>±0.02 tons</td></tr> <tr> <td>Calibration Date:</td><td>20.10.2020 23.11.2023</td></tr> <tr> <td>Calibration Standard:</td><td>The Weighing Instrument Regulations⁵⁹</td></tr> </table>	Name:	Weighbridge	Manufacturer:	ESİT	Type:	Non-Automatic weighbridge with a capacity of 60,000-80,000 kg	Accuracy Class:	±0.02 tons	Calibration Date:	20.10.2020 23.11.2023	Calibration Standard:	The Weighing Instrument Regulations ⁵⁹
Name:	Weighbridge												
Manufacturer:	ESİT												
Type:	Non-Automatic weighbridge with a capacity of 60,000-80,000 kg												
Accuracy Class:	±0.02 tons												
Calibration Date:	20.10.2020 23.11.2023												
Calibration Standard:	The Weighing Instrument Regulations ⁵⁹												
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested by the manufacturer in compliance with the Turkish legislation titled 'The Weighing Instrument Regulations.												
Purpose of data	Used in Calculation of baseline emissions												
Calculation method	The weighbridge data monitored to measure quantity of raw manure (Q _y) supplied by each farm is multiplied with DM ratios for each farm in order to calculate quantity of manure in dry basis.												
Comments	-												
Data / Parameter	Q _y												
Data unit	Tonnes-wet basis/year												

⁵⁹

[https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=18793&MevzuatTur=7&MevzuatTertip=5#:~:text=a\)%20Tart%C4%B1%20aletlerinin%20periyodik%20muayene,y%C4%B1%20birinci%20y%C4%B1%20olarak%20say%C4%B1%C4%B1r.](https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=18793&MevzuatTur=7&MevzuatTertip=5#:~:text=a)%20Tart%C4%B1%20aletlerinin%20periyodik%20muayene,y%C4%B1%20birinci%20y%C4%B1%20olarak%20say%C4%B1%C4%B1r.) (Accessed on 08/12/2023)

Description	Quantity of raw manure co digested in the year y																								
Source of data	On site measurement, operational logbooks (weighbridge records)																								
Description of measurement methods and procedures to be applied	Each manure load transported to the site is measured with weight balance / or truck load capacity. (In case of weigh bridge failure, rated capacity of truck load is considered for the calculation.)																								
Frequency of monitoring/recording	Annually, based on daily measurement and monthly aggregation																								
Value monitored	<table><tr><th>Farm Name</th><th>Q_y (tonnes/yr)</th></tr><tr><td>Ali Bey Çiftliği (Necmettin Erden)</td><td>7,448</td></tr><tr><td>Erol Yılmaz</td><td>7,865</td></tr><tr><td>Bey Besi (Veysel Karabey)</td><td>8,750</td></tr><tr><td>Şenol Yılmaz Besi Çiftliği</td><td>3,650</td></tr><tr><td>Saygın Bey (Veysel Karabey)</td><td>15,865</td></tr><tr><td>Aytekin Besi</td><td>5,738</td></tr><tr><td>has besi</td><td>37,834</td></tr><tr><td>izzet alp</td><td>12,587</td></tr><tr><td>yılmazlar besi (kerim yılmaz)</td><td>12,693</td></tr><tr><td>TOTAL</td><td>112,430</td></tr></table>			Farm Name	Q _y (tonnes/yr)	Ali Bey Çiftliği (Necmettin Erden)	7,448	Erol Yılmaz	7,865	Bey Besi (Veysel Karabey)	8,750	Şenol Yılmaz Besi Çiftliği	3,650	Saygın Bey (Veysel Karabey)	15,865	Aytekin Besi	5,738	has besi	37,834	izzet alp	12,587	yılmazlar besi (kerim yılmaz)	12,693	TOTAL	112,430
Farm Name	Q _y (tonnes/yr)																								
Ali Bey Çiftliği (Necmettin Erden)	7,448																								
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TOTAL	112,430																								
Monitoring equipment	Weighbridge scales at the entrance and/or truck load capacity and trip aggregation.																								

	<table> <tr> <td>Name:</td><td>Weighbridge</td></tr> <tr> <td>Manufacturer:</td><td>ESİT</td></tr> <tr> <td>Type:</td><td>Non-Automatic weighbridge with a capacity of 60,000-80,000 kg</td></tr> <tr> <td>Accuracy Class:</td><td>±0.02 tons</td></tr> <tr> <td>Calibration Date:</td><td>20.10.2020 23.11.2023</td></tr> <tr> <td>Calibration Standard:</td><td>The Weighing Instrument Regulations⁶⁰</td></tr> </table>	Name:	Weighbridge	Manufacturer:	ESİT	Type:	Non-Automatic weighbridge with a capacity of 60,000-80,000 kg	Accuracy Class:	±0.02 tons	Calibration Date:	20.10.2020 23.11.2023	Calibration Standard:	The Weighing Instrument Regulations ⁶⁰
Name:	Weighbridge												
Manufacturer:	ESİT												
Type:	Non-Automatic weighbridge with a capacity of 60,000-80,000 kg												
Accuracy Class:	±0.02 tons												
Calibration Date:	20.10.2020 23.11.2023												
Calibration Standard:	The Weighing Instrument Regulations ⁶⁰												
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested by the manufacturer in compliance with the Turkish legislation titled 'The Weighing Instrument Regulations'.												
Purpose of data	Used in Calculation of project emissions												
Calculation method	-												
Comments	-												

Data / Parameter	SVS _{LT,y}
Data unit	tonnes VS/tonnes-dm
Description	Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y
Source of data	Project proponents, Lab logbook records
Description of measurement methods and procedures to be applied	Volatile solids are measured/determined at the laboratory of the project as per Annex 2 of the methodology AM00073. These weekly measurements are then averaged for each farm to find the SVS value on the monitoring year.
Frequency of monitoring/recording	Weekly, aggregated to find annually.

⁶⁰

[https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=18793&MevzuatTur=7&MevzuatTertip=5#:~:text=a\)%20Tart%C4%B1%20aletlerin%20periyodik%20muayene,y%C4%B1%20birinci%20y%C4%B1%20olarak%20say%C4%B1%C4%B1r.](https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=18793&MevzuatTur=7&MevzuatTertip=5#:~:text=a)%20Tart%C4%B1%20aletlerin%20periyodik%20muayene,y%C4%B1%20birinci%20y%C4%B1%20olarak%20say%C4%B1%C4%B1r.) (Accessed on 08/12/2023)

Value monitored		<table><tr><th>Farm Name</th><th>SVS_{LT,y} (tonnes VS /tonnes-dm)</th></tr><tr><td>Ali Bey Çiftliği (Necmettin Erden)</td><td>0.80</td></tr><tr><td>Erol Yılmaz</td><td>0.82</td></tr><tr><td>Bey Besi (Veysel Karabey)</td><td>0.75</td></tr><tr><td>Şenol Yılmaz Besi Çiftliği</td><td>0.76</td></tr><tr><td>Saygın Bey (Veysel Karabey)</td><td>0.7</td></tr><tr><td>Aytekin Besi</td><td>0.68</td></tr><tr><td>has besi</td><td>0.72</td></tr><tr><td>izzet alp</td><td>0.76</td></tr><tr><td>yılmazlar besi (kerim yılmaz)</td><td>0.78</td></tr></table>	Farm Name	SVS _{LT,y} (tonnes VS /tonnes-dm)	Ali Bey Çiftliği (Necmettin Erden)	0.80	Erol Yılmaz	0.82	Bey Besi (Veysel Karabey)	0.75	Şenol Yılmaz Besi Çiftliği	0.76	Saygın Bey (Veysel Karabey)	0.7	Aytekin Besi	0.68	has besi	0.72	izzet alp	0.76	yılmazlar besi (kerim yılmaz)	0.78
	Farm Name	SVS _{LT,y} (tonnes VS /tonnes-dm)																				
	Ali Bey Çiftliği (Necmettin Erden)	0.80																				
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	Saygın Bey (Veysel Karabey)	0.7																				
	Aytekin Besi	0.68																				
	has besi	0.72																				
	izzet alp	0.76																				
	yılmazlar besi (kerim yılmaz)	0.78																				
Monitoring equipment	The three main equipment used to determine volatile solid excretion rate in animal waste are as follows:																					
	- Muffle Furnace: NASTECH NST-1100-40-E4 - Dry air sterilizer: Nükleon NST Series; - Precision balance: OHAUS PR224 Series																					
	<table><tr><td>Name:</td><td>Muffle Furnace</td></tr><tr><td>Type:</td><td>NASTECH NST-1100-40-E4</td></tr><tr><td>Serial Number:</td><td>E308-201217-05</td></tr><tr><td>Accuracy Class:</td><td>≤ +/- 0.5%</td></tr><tr><td>Calibration Date:</td><td>25.10.2020</td></tr><tr><td>Calibration Standard:</td><td>Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change</td></tr></table>	Name:	Muffle Furnace	Type:	NASTECH NST-1100-40-E4	Serial Number:	E308-201217-05	Accuracy Class:	≤ +/- 0.5%	Calibration Date:	25.10.2020	Calibration Standard:	Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change									
Name:	Muffle Furnace																					
Type:	NASTECH NST-1100-40-E4																					
Serial Number:	E308-201217-05																					
Accuracy Class:	≤ +/- 0.5%																					
Calibration Date:	25.10.2020																					
Calibration Standard:	Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change																					

	Name:	Dry air sterilizer
	Type:	NÜKLEON NST Series / NST 55
	Serial Number:	20ST55213
	Accuracy Class:	≤ +/- 0.5%
	Calibration Date:	25.10.2020
	Calibration Standard:	Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change
	Name:	Scale
	Type:	OHAUS PR Series PR224
	Serial Number:	B948669554
	Accuracy Class:	≤ +/- 0.5%
	Calibration Date:	25.10.2020
	Calibration Standard:	Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 25.10.2020 by Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change ⁶¹ .	
Purpose of the data	Used in Calculation of baseline emissions	
Calculation method	Analyses are performed in line with the Annex 2: 'Method for determination for Volatile Solids in animal waste' and the Annex 4: 'Guidance on sample extraction and statistical procedures' of the approved baseline and monitoring CDM methodology of AM0073: "GHG emission reductions through multi-site manure collection and treatment in a central plant, Version 1.0" ⁶² . After determining the weekly VS values as mentioned above, their yearly average is calculated for each farm to obtain SVS _{LT,y} .	
Comments	Detailed lab logbook data sheet on W1, W2 and Wf used in the	

⁶¹ https://webdosya.csb.gov.tr/db/lab/editordosya/cevre_laboratuvarlari_denetim_kilavuzu.pdf (Accessed on 10/11/2022)

⁶² <https://cdm.unfccc.int/methodologies/DB/2N19WQ6DCXNYRNJVZQQOHG7TK0Q2D8> (Accessed on 08/12/2023)

	SVS calculations as per Annex 2 of the applied Meth. are recorded in the lab logbook in hard copy version.
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Data / Parameter	B _{0,LT}
Data unit	m ³ CH ₄ /kg-dm
Description	Maximum methane producing potential of the volatile solid generated for animal type 'LT'
Source of data	2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019, as updated in July 2023), Volume 4, Chapter 10, Table 10.16A (Updated), page 10.67
Description of measurement methods and procedures to be applied	In accordance with 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019, as updated in July 2023), Turkey is regarded under Middle East region.
Frequency of monitoring/recording	Annually
Value applied	For Other Region – High Productivity Systems (PS), Non-Dairy Cattle: 0.18;
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Used in Calculation of baseline emissions
Comments	Since there is no country specific data on B_{0,LT} values for dairy cattle, non-dairy cattle and hens, default values are used for developed countries in line with IPCC 2019 Refinement to the IPCC 2006 Guidelines, as updated in July 2023 (page 10.67 Table 10.16A-Updated). Considering that only one type of animal manure (non-dairy) is used as feedstock in the biogas production process of the

	project, the default maximum methane producing potential of the volatile solid generated values non-dairy cattle in the animal category have been chosen. According to 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019, as updated in July 2023), since Turkey is considered a Middle East country, the above-mentioned values (Non-Dairy Cattle: 0.18) in the other region category have been used. Since the farms supplying feedstocks to the project are doing commercial production and sales-oriented farms, high-productivity values for other regions have been chosen as per the applied methodology (ibid.: 10.13).
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Data / Parameter	CT _y										
Data unit	tonnes/truck										
Description	Average truck capacity for transportation										
Source of data	On site measurement										
Description of measurement methods and procedures to be applied	Minimum capacity of the special trucks stated in the licences is taken as a conservative value.										
Frequency of monitoring/recording	-										
Value monitored	21 tonnes/truck										
Monitoring equipment	Weighbridge and vehicle licences <table border="1"> <tr> <td>Name:</td><td>Weighbridge</td></tr> <tr> <td>Manufacturer:</td><td>ESIT</td></tr> <tr> <td>Type:</td><td>Non-Automatic weighbridge with a capacity of 60,000-80,000 kg</td></tr> <tr> <td>Accuracy Class:</td><td>±0.02 tons</td></tr> <tr> <td>Calibration Date:</td><td>20.10.2020</td></tr> </table>	Name:	Weighbridge	Manufacturer:	ESIT	Type:	Non-Automatic weighbridge with a capacity of 60,000-80,000 kg	Accuracy Class:	±0.02 tons	Calibration Date:	20.10.2020
Name:	Weighbridge										
Manufacturer:	ESIT										
Type:	Non-Automatic weighbridge with a capacity of 60,000-80,000 kg										
Accuracy Class:	±0.02 tons										
Calibration Date:	20.10.2020										

	23.11.2023
Calibration Standard:	The Weighing Instrument Regulations ⁶³
QA/QC procedures to be applied	Values are monitored via the recordings of the net capacity of the trucks at weighbridge readings and cross-checked with the vehicle licenses.
Purpose of data	Used in Calculation of project emissions
Calculation method	-
Comments	The parameter is also used to define the parameter $EF_{CO_2,transport}$ as per the Methodological Tool: Project and leakage emissions from transportation of freight, Version 01.0.0

Data / Parameter	DAF _w
Data unit	km/truck
Description	Average incremental distance for raw solid manure transportation
Source of data	Project proponent measurements of distance
Description of measurement methods and procedures to be applied	Direct measurement of the distance between the collection point of manure-to-manure discharge point
Frequency of monitoring/recording	A fixed value is assumed until the locations of the collection point and discharge point are changed.
Value monitored	14.3 km/truck
Monitoring equipment	Google maps
QA/QC procedures to be applied	-

⁶³

[https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=18793&MevzuatTur=7&MevzuatTertip=5#:~:text=a\)%20Tart%C4%B1%20aletlerin%20periyodik%20muayene,y%C4%B1%20birinci%20y%C4%B1%20olarak%20say%C4%B1%C4%B1r.](https://www.mevzuat.gov.tr/mevzuat?MevzuatNo=18793&MevzuatTur=7&MevzuatTertip=5#:~:text=a)%20Tart%C4%B1%20aletlerin%20periyodik%20muayene,y%C4%B1%20birinci%20y%C4%B1%20olarak%20say%C4%B1%C4%B1r.) (Accessed on 08/12/2023)

Purpose of data	Used in Calculation of project emissions
Calculation method	-
Comments	In case of any changes in the source of manure it will be logged and recorded electronically for the crediting period plus two years.

Data / Parameter	$EG_{PJ,y} = EG_{facility,y}$																				
Data unit	MWh/yr																				
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y																				
Source of data	Electricity meter recordings / Bills issued to EPIAŞ.																				
Description of measurement methods and procedures to be applied	This parameter is monitored electricity meter Class 0.5S																				
Frequency of monitoring/recording	Continuous measurements and monthly recorded.																				
Value applied	6,658																				
Monitoring equipment	Electricity meters <table border="1"> <tr> <td>Name:</td><td>Electricity meter – Back up</td></tr> <tr> <td>Type:</td><td>MAKEL C520.AMT.2556</td></tr> <tr> <td>Accuracy Class:</td><td>0.5S</td></tr> <tr> <td>Series Number:</td><td>80281425</td></tr> <tr> <td>Date of Calibrations:</td><td>05.10.2020</td></tr> <tr> <td>Calibration Standard:</td><td>The Measurement and Measuring Instrument Regulations</td></tr> </table> <table border="1"> <tr> <td>Name:</td><td>Electricity meter – Main</td></tr> <tr> <td>Type:</td><td>MAKEL C520.AMT.2556</td></tr> <tr> <td>Accuracy Class:</td><td>0.5S</td></tr> <tr> <td>Series Number:</td><td>802720060</td></tr> </table>	Name:	Electricity meter – Back up	Type:	MAKEL C520.AMT.2556	Accuracy Class:	0.5S	Series Number:	80281425	Date of Calibrations:	05.10.2020	Calibration Standard:	The Measurement and Measuring Instrument Regulations	Name:	Electricity meter – Main	Type:	MAKEL C520.AMT.2556	Accuracy Class:	0.5S	Series Number:	802720060
Name:	Electricity meter – Back up																				
Type:	MAKEL C520.AMT.2556																				
Accuracy Class:	0.5S																				
Series Number:	80281425																				
Date of Calibrations:	05.10.2020																				
Calibration Standard:	The Measurement and Measuring Instrument Regulations																				
Name:	Electricity meter – Main																				
Type:	MAKEL C520.AMT.2556																				
Accuracy Class:	0.5S																				
Series Number:	802720060																				

	Date of Calibrations:	10.10.2020
	Calibration Standard:	The Measurement and Measuring Instrument Regulations
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 05.10.2020 and 10.10.2020 both by the manufacturer and the officially designated company in compliance with the Turkish legislation titled 'The Measurement and Measuring Instrument Regulations' ⁶⁴ , and data was recorded and backed-up.	
Purpose of data	Used in Calculation of project emissions	
Calculation method	-	
Comments	-	

Data / Parameter	EC _{PJ,j,y}
Data unit	MWh/yr
Description	Quantity of electricity consumed by the project electricity consumption source the national grid in year y
Source of data	Bill issued by regional electricity distribution company
Description of measurement methods and procedures to be applied	This parameter is monitored via electricity meter Class 0.5S
Frequency of monitoring/recording	Monitoring: Continuous measurements and monthly recorded Recording: Archive electronically during project plus 5 years
Value monitored	0

⁶⁴

<https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5#:~:text=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1l%C4%B1r>

Monitoring equipment	Electricity meters	
	Name:	Electricity meter – Back up
	Type:	MAKEL C520.AMT.2556
	Accuracy Class:	0.5S
	Series Number:	80281425
	Date of Calibrations:	05.10.2020
	Calibration Standard:	'The Measurement and Measuring Instrument Regulations'
	Name:	Electricity meter – Main
	Type:	MAKEL C520.AMT.2556
QA/QC procedures to be applied	Accuracy Class:	0.5S
	Series Number:	802720060
	Date of Calibrations:	10.10.2020
	Calibration Standard:	'The Measurement and Measuring Instrument Regulations'
Purpose of data	The monitoring equipment was calibrated and tested on 05.10.2020 and 10.10.2020 both by the manufacturer and the officially designated company in compliance with the Turkish legislation titled 'The Measurement and Measuring Instrument Regulations' ⁶⁵ , and data was recorded and backed-up.	
Calculation method	Used in	
	Calculation of project emissions	
Comments	-	
	In case the electricity consumes from the national grid due to any circumstance, such as emergency, the relevant calculations will be included in the upcoming monitoring period.	

⁶⁵

<https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5#:~:te xt=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1l%C4%B1r> (Accessed on 08/12/2023)

Data / Parameter	$EF_{EF,j,y} = EF_{grid,y}$
Data unit	tCO _{2e} /MWh
Description	Emission factor for the Turkish National Grid
Source of data	Published data from The Ministry of the Energy and Natural Resources in Turkey
Description of measurement methods and procedures to be applied	Obtained from the most recent official national grid emission factor data ⁶⁶ published by the Ministry of the Energy and Natural Resources in Turkey in time of the project's verification
Frequency of monitoring/recording	Annually
Value monitored	0.5552 ⁶⁷
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Used in • Calculation of baseline emissions • Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	TDL _{j,y}
Data unit	-

⁶⁶ It is value that obtained in time of validation and obtained from the most recent official national grid emission factor data published by the Ministry of the Energy and Natural Resources in Turkey. This value might be subject to a change in time of verification in accordance with the revisions of the Ministry of the Energy and Natural Resources in Turkey.

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUESEmisyonFktr/Belgeler/Bform2020.pdf> (Accessed on 08/12/2023)

⁶⁷ It is value that obtained in time of validation.

Description	Average technical transmission and distribution losses for providing electricity to source j in year y
Source of data	Calculated from the Data Provided by Turkish Electricity Transmission Corporation (TEİAŞ) ⁶⁸ in accordance with Methodological Tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, Version 3.0 ⁶⁹ .
Description of measurement methods and procedures to be applied	The annual average value based on the most recent data available within Turkey is used from the data provided by TEİAŞ. In accordance with the Methodological Tool, for the data from the relevant year, 2021, is absence, most recent figures are used, but not older than 5 years. Accordingly, the annual network losses for the years of 2017, 2018, and 2019 are taken into account in calculating $TDL_{j,y}$.
Frequency of monitoring/recording	yearly
Value monitored	0.11 ⁷⁰
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	Used in Calculation of project emissions
Calculation method	-
Comments	-

Data / Parameter	$Q_{biogas,y}$
Data unit	Nm ³ biogas
Description	Amount of biogas collected at the digester outlet in year y

⁶⁸ <https://webapi.teias.gov.tr/file/512cbf1d-0ca3-4492-b901-3722c7b682f7?download> (Accessed on 08/12/2023)

⁶⁹ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf> (Accessed on 08/12/2023)

⁷⁰ It is value that obtained in time of validation.

Source of data	Flow meter readings												
Description of measurement methods and procedures to be applied	The Scada software system which is an automated control panel providing data on biogas production processes at the site instantaneously, provides the data on the amount of biogas produced in the anaerobic digesters, along with the methane content in the biogas. These data are exported daily and recorded to the analyser's logbook.												
Frequency of monitoring/recording	Continuously measurement by the flow meter, aggregated monthly and yearly.												
Value monitored	3,197,940 ⁷¹												
Monitoring equipment	The equipment used to determine quantity of methane produced in the anaerobic digester is as follows: Gas flow meter: <table border="1"> <tr> <td>Name:</td><td>Flow Meter - 1</td></tr> <tr> <td>Type:</td><td>PROSONIC FLOW B200</td></tr> <tr> <td>Accuracy Class:</td><td>≤ +/- 0.2%</td></tr> <tr> <td>Series Number:</td><td>R8107702000</td></tr> <tr> <td>Calibration Date:</td><td>31.08.2020</td></tr> <tr> <td>Calibration Standard:</td><td>The Measurement and Measuring Instrument Regulations⁷²</td></tr> </table>	Name:	Flow Meter - 1	Type:	PROSONIC FLOW B200	Accuracy Class:	≤ +/- 0.2%	Series Number:	R8107702000	Calibration Date:	31.08.2020	Calibration Standard:	The Measurement and Measuring Instrument Regulations ⁷²
Name:	Flow Meter - 1												
Type:	PROSONIC FLOW B200												
Accuracy Class:	≤ +/- 0.2%												
Series Number:	R8107702000												
Calibration Date:	31.08.2020												
Calibration Standard:	The Measurement and Measuring Instrument Regulations ⁷²												
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 31.08.2020 by the manufacturer in compliance with the Turkish legislation titled 'The Measurement and Measuring Instrument Regulations' ⁷² .												
Purpose of data	Used in Calculation of project emissions												
Calculation method	-												

⁷¹ The representative one-year value was obtained from Flow meter - 1 installed at the biogas plant for the estimation of this parameter in time of validation.

⁷²

<https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5#:~:te xt=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1l%C4%B1r> (Accessed on 08/12/2023)

Comments

-

Data / Parameter	$Q_{CH_4,y}$												
Data unit	tCH ₄ /y												
Description	Quantity of methane produced in the anaerobic digester in year y (tCH ₄ /y)												
Source of data	Project developer records												
Description of measurement methods and procedures to be applied	The SCADA software system which is an automated control panel providing data on biogas production processes at the site instantaneously, provides the data on the amount of biogas produced in the anaerobic digesters, along with the methane content in the biogas. These data are exported daily and recorded to the analyser's logbook.												
Frequency of monitoring/recording	Annually, based on daily records, monthly aggregation												
Value monitored	1,286												
Monitoring equipment	The main equipment used to determine quantity of methane produced in the anaerobic digester is as follows: Gas flow meter: Proline Prosonic Flow B 200 <table border="1"> <tr> <td>Name:</td><td>Flow Meter – 1</td></tr> <tr> <td>Type:</td><td>PROSONIC FLOW B200</td></tr> <tr> <td>Accuracy Class:</td><td>≤ +/- 0.2%</td></tr> <tr> <td>Series Number:</td><td>R8107702000</td></tr> <tr> <td>Calibration Date:</td><td>31.08.2020</td></tr> <tr> <td>Calibration Standard:</td><td>The Measurement and Measuring Instrument Regulations'</td></tr> </table>	Name:	Flow Meter – 1	Type:	PROSONIC FLOW B200	Accuracy Class:	≤ +/- 0.2%	Series Number:	R8107702000	Calibration Date:	31.08.2020	Calibration Standard:	The Measurement and Measuring Instrument Regulations'
Name:	Flow Meter – 1												
Type:	PROSONIC FLOW B200												
Accuracy Class:	≤ +/- 0.2%												
Series Number:	R8107702000												
Calibration Date:	31.08.2020												
Calibration Standard:	The Measurement and Measuring Instrument Regulations'												
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 31.08.2020 by the manufacturer in compliance with the Turkish legislation titled 'The Measurement and Measuring Instrument Regulations' ⁷³ .												

Purpose of data	Used in Calculation of project emissions
Calculation method	This parameter is calculated using the formula below in accordance with the Tool 14: Project and leakage emissions from anaerobic digesters as per the applied methodology. $Q_{CH_4} = f_{CH_4, default} \times \rho_{CH_4}$
Comments	-

Data / Parameter	MCF_j
Data unit	Fraction
Description	Annual methane conversion factor (MCF) for the baseline animal manure management system j
Source of data	Obtained from 2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories (IPCC 2019, as updated in July 2023), Volume 4, Chapter 10, Table 10.17 (Updated), page 68
Frequency of monitoring/recording	Annually
Value applied:	0.76
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of Data	Used in Calculation of baseline emissions
Calculation method	<u>During these nine months:</u> Mean annual temperature is 12°C⁷⁴; Mean annual precipitation (MAP) is 391.1⁷⁵;

xt=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1l%C4%B1r (Accessed on 08/12/2023)

⁷⁴ <https://mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=ANKARA> (Accessed on 08/12/2023)

	Potential Evapotranspiration (PET) is 1,184.2⁷⁶. As per the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, as updated in July 2023, Volume 4: Agriculture, Forestry and Other Land Use, Chapter 3: Consistent Representation of Lands, Figure 3A.5.2, page 3.48, considering the meteorological data (mean annual temperature and MAP/PET) stated above, ANKARA's climate zone is determined as Warm Temperature Dry. Furthermore, in accordance with Information Required to Determine Climate Zones According to Chapter 10 of Volume 4 Current Guideline of IPCC 2019, as updated in July 2023 page 10.127, Ankara is considered as Warm Temperature Dry as its climate zone as per its annual average temperature. J is Uncovered anaerobic lagoon in the baseline scenario. Thus, $MCF_{\text{Uncovered anaerobic lagoon}}$ is accounted as 0.76.
Comments	In time of validation, PP has used the official public data and EIA reports to define 'Mean Annual Temperature', 'MAP', and 'PET', parameters necessary to identify the project site's climate, in line with the IPCC Guidelines. However, during the initial monitoring processes, PPs are going to obtain these data, such as the parameter 'T' from the nearest meteorology center to the project location. Hence, MCF_j parameter may be subjected to changes during the monitoring period in line with the changes in climate where the project is located at. Also, in the calculation of annual baseline methane emissions generated at the farm, this value is multiplied by a conservativeness factor being 0.94 that is taken.

Data / Parameter	DM
Data unit	%
Description	Residue remaining after water is removed from waste material by evaporation; dry matter

⁷⁵ <https://mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=ANKARA> (Accessed on 08/12/2023)

⁷⁶ <https://eced.csb.gov.tr/jsp/ek1/14501#> (page 83) (Accessed on 08/12/2023)

Source of data	Lab logbook records																					
Description of measurement methods and procedures to be applied	Measurements are made with Shimadzu UniBloc MOC63u moisture analyser.																					
Frequency of monitoring/recording	Weekly, aggregated to find annually.																					
Value monitored	<table><thead><tr><th>Farm Name</th><th>DM %</th></tr></thead><tbody><tr><td>Ali Bey Çiftliği (Necmettin Erden)</td><td>0.16</td></tr><tr><td>Erol Yılmaz</td><td>0.13</td></tr><tr><td>Bey Besi (Veysel Karabey)</td><td>0.14</td></tr><tr><td>Şenol Yılmaz Besi Çiftliği</td><td>0.08</td></tr><tr><td>Saygın Bey (Veysel Karabey)</td><td>0.11</td></tr><tr><td>Aytekin Besi</td><td>0.12</td></tr><tr><td>has besi</td><td>0.14</td></tr><tr><td>izzet alp</td><td>0.24</td></tr><tr><td>yılmazlar besi (kerim yılmaz)</td><td>0.18</td></tr></tbody></table>		Farm Name	DM %	Ali Bey Çiftliği (Necmettin Erden)	0.16	Erol Yılmaz	0.13	Bey Besi (Veysel Karabey)	0.14	Şenol Yılmaz Besi Çiftliği	0.08	Saygın Bey (Veysel Karabey)	0.11	Aytekin Besi	0.12	has besi	0.14	izzet alp	0.24	yılmazlar besi (kerim yılmaz)	0.18
Farm Name	DM %																					
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has besi	0.14																					
izzet alp	0.24																					
yılmazlar besi (kerim yılmaz)	0.18																					
Monitoring equipment	Moisture analyzer: SHIMADZU MOC63u <table><tr><td>Name:</td><td>Moisture analyzer</td></tr><tr><td>Type:</td><td>SHIMADZU MOC63u</td></tr><tr><td>Accuracy Class:</td><td>≤ +/- 0.5%</td></tr><tr><td>Calibration Date:</td><td>25.10.2020</td></tr><tr><td>Calibration Standard:</td><td>Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization</td></tr></table>		Name:	Moisture analyzer	Type:	SHIMADZU MOC63u	Accuracy Class:	≤ +/- 0.5%	Calibration Date:	25.10.2020	Calibration Standard:	Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization										
Name:	Moisture analyzer																					
Type:	SHIMADZU MOC63u																					
Accuracy Class:	≤ +/- 0.5%																					
Calibration Date:	25.10.2020																					
Calibration Standard:	Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization																					

		and Climate Change
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 25.10.2020 by Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change.	
Purpose of data	Used in Calculation of baseline emissions	
Calculation method	-	
Comments	The moisture analyzer initially measures the weight of the wet sample, then heats it up until it's dry, and measures the dried sample weight to find the dry matter fraction. The heating procedure continues until the weight reading is no longer changing. The device in the end gives the DM fraction on its digital screen, hence this value is taken into account	

Data / Parameter	$\eta_{flare,m}$	
Data unit	%	
Description	The flare efficiency	
Source of data	Determined based on the applied Methodology AMS-III.D paragraph 22	
Description of measurement methods and procedures to be applied	This rationale is based on the Paragraph 22 of AMS-III.D., stating "If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed."	
Frequency of monitoring/recording	-	
Value monitored	-	
Monitoring equipment	Flow meter – 2	
	Name:	Flow Meter - 2

	Type:	PROSONIC FLOW B200
	Accuracy Class:	≤ +/- 0.2%
	Series Number:	R8106902000
	Calibration Date:	26.08.2020
	Calibration Standard:	The Measurement and Measuring Instrument Regulations
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 26.08.2020 by the manufacturer in compliance with the Turkish legislation titled 'The Measurement and Measuring Instrument Regulations' ⁷⁷ .	
Purpose of data	Used in Calculation of project emissions	
Calculation method	-	
Comments		

Data / Parameter	EGy
Data unit	MWh
Description	Total electricity generated from the recovered biogas in year y
Source of data	Electricity meter recordings
Description of measurement methods and procedures to be applied	This parameter is monitored electricity meter Class 0.5S
Frequency of monitoring/recording	Continuous measurements.
Value applied	-
Monitoring equipment	Electricity meters

⁷⁷

<https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5#:~:te xt=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1l%C4%B1r> (Accessed on 08/12/2023)

	Name:	Electricity meter – Back up
	Type:	MAKEL C520.AMT.2556
	Accuracy Class:	0.5S
	Series Number:	80281425
	Date of Calibrations:	05.10.2020
	Calibration Standard:	The Measurement and Measuring Instrument Regulations
	Name:	Electricity meter – Main
	Type:	MAKEL C520.AMT.2556
	Accuracy Class:	0.5S
	Series Number:	802720060
	Date of Calibrations:	10.10.2020
	Calibration Standard:	The Measurement and Measuring Instrument Regulations
QA/QC procedures to be applied	The monitoring equipment was calibrated and tested on 05.10.2020 and 10.10.2020 both by the manufacturer and the officially designated company in compliance with the Turkish legislation titled ‘The Measurement and Measuring Instrument Regulations’ ⁷⁸ , and data was recorded and backed-up.	
Purpose of data	Used in Calculation of ex-post net emission reductions	
Calculation method	-	
Comments	-	

Data / Parameter	EE _y
Data unit	%
Description	Energy conversion efficiency of the project equipment

⁷⁸

<https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5#:~:te xt=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1l%C4%B1r>

Source of data	Determined based on the applied Methodology AMS-III.D paragraph 30
Description of measurement methods and procedures to be applied	Default efficiency of 40% is adopted as per para 30 of the applied Methodology
Frequency of monitoring/recording	-
Value monitored	40%
Monitoring equipment	- -
QA/QC procedures to be applied	
Purpose of data	Used in Calculation of net emission reduction
Calculation method	-
Comments	This parameter will be used for ex-post emission reduction calculations in the upcoming monitoring periods.

5.3 Monitoring Plan

Organizational Structure

The Carbon Manager assigned to the project, is responsible for designing the monitoring activities as per the guidelines of the Methodology. This way, carbon manager ensures the real, reliable, and traceable measurement of the GHG emission reductions. The organizational structure of the project is given in Figure 28.

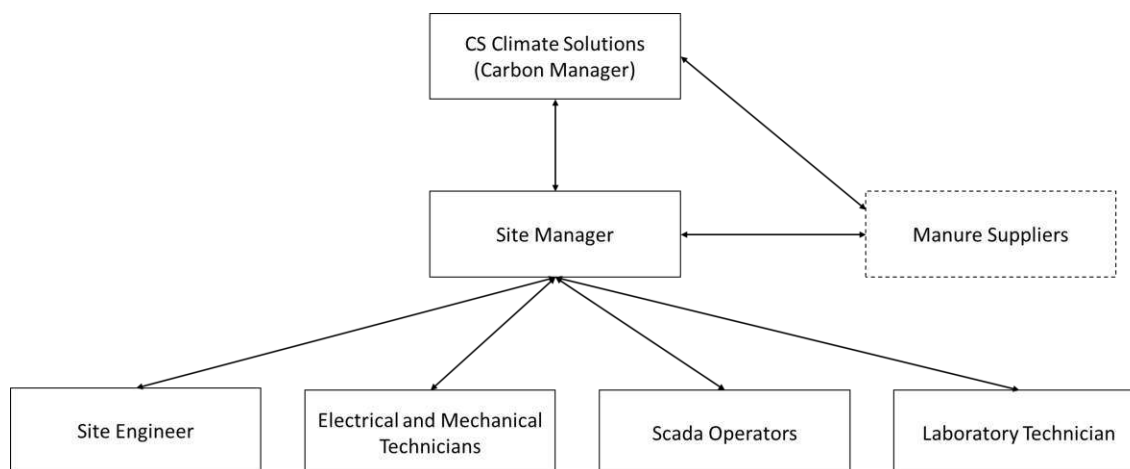


Figure 28. Organizational Structure of the Plant

The key roles are defined as follows:

Carbon manager – developing the project, determining the monitoring procedure, and providing the necessary guidance to the site manager as well as to the site personnel. Besides, carbon manager also prepares monitoring surveys to obtain information regarding manure suppliers such as the monitoring parameters N_p (number of animals produced annually of type LT) and N_{da} (number of days animal is alive in the farm in the year), and as well to obtain information regarding the situation before the project activity. The appropriateness of monitoring surveys that are conducted by the site engineer is checked by the carbon manager. Lastly, carbon manager ensures the monitoring plan is implemented in accordance with the methodological requirements.

Site manager – to control all activities and operations (managing the budget and personnel, process controlling) in the plant, including the control unit of the plant, monitoring equipment and the staff. The site manager records the amount of diesel oil used by vehicles used to transport manure daily. Another responsibility of the site manager is to carry out monitoring surveys with the manure supplying farms. Surveys are kept in hard copy formats at the project site, then, passed to electronic records. Overall, all the collected monitoring data from the site personnel, as well as the manure suppliers, are conveyed to the Carbon Manager by the site manager.

Site engineer – to control biogas production process in the plant – control of process parameters, following up lab tests, comment on lab analysis, biogas measurements and manure management, and ensuring the calibration of the equipment. Site engineer is also responsible for gathering the weighbridge recordings from the assigned personnel of the plant. In addition, site engineer is in charge of reporting to the site manager.

Electrical and mechanical technicians – the continuous operation of the gas engines, all electrical and mechanical equipment for electricity generation, as well as maintenance and calibration. Responsible of keeping operational logbook records and submitting them to the site manager periodically.

Both the site manager and the electrical and mechanical technicians are in charge of recording the data on amount of electricity generated and supplied to the national grid

SCADA¹⁰² operator– monitors all equipment and process variables (flow rate, biogas content, temperature, pressure, etc.) on Scada System. Keeps electronic records extracted from Scada and archives them.

Both the electrical/mechanical technicians and SCADA operator at site is responsible to record the data from flow meter as well as flame detector associated to the flare.

Laboratory personnel– accredited technicians responsible for collecting samples from the assigned sampling points and conducting the necessary measurements/analysis on a periodical basis. The analyses results are, first, recorded by the personnel to lab logbooks in a hard copy format and then, to electronic records. The data recorded electronically is going to be archived during project plus five years as per the applied Meth. The lab personnel is also in charge of ensuring the calibration of the lab equipment periodically.

Description of the Monitoring Plan

The table below describes the parameters monitored at the project site. Also, the following figure shows on the flow sheet at which part of the project activity those monitoring parameters are measured.

Table 6. Parameters monitored at the project site

Monitoring Parameter	Method to monitor/record	Responsible
EG _{facility,y} - Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	Within the project activity, two electricity meters are installed for the monitoring of net electricity amount exported to the national grid. The electricity meter readings are recorded	Site manager / the electrical and mechanical technicians

¹⁰²

<https://search.abb.com/library/Download.aspx?DocumentID=3BSE069330&LanguageCode=en&DocumentPartId=&Action=Launch>

	daily by SCADA system. Also, net electricity exports are cross-checked via electricity sales invoices which are recorded by the site manager. The electricity meters are calibrated by the officially accredited entity in line with the Turkish regulations. Collected data is archived in both hard and soft copy format. Please refer to Section 4.2 and Section 5.2 in VCS PD.	
$EC_{PJ,j,y}$ – Quantity of electricity consumed by the project electricity consumption source the national grid in year y	Within the project activity, two electricity meters are installed for the monitoring of electricity consumption. The electricity meter readings are recorded daily by SCADA system. Also, electricity consumption from the grid is cross-checked via electricity invoices issued by regional electricity distribution company which are recorded by site manager. The electricity meters are calibrated by the officially accredited entity in line with the Turkish regulations. Collected data is archived in both hard and soft copy format. Please refer to Section 4.2 and 5.2 in VCS PD.	Site manager / the electrical and mechanical technicians / regional electricity distribution company
$Q_{CH4,y}$ - Quantity of methane produced in the anaerobic digester in year y	Within the project activity, there is one biogas flow meter (Flow Meter – 2) installed at the combined biogas lines for the monitoring of capture biogas.	SCADA operator / Site engineer

	This parameter is measured continuously by the biogas flow meters and recorded through the SCADA system. Continuous recordings of biogas flow can be exported from the SCADA system. Accumulated data for daily and monthly records are exported by the SCADA operator. The amount of biogas generated is archived in hard and soft copy format. The relevant flow meters are calibrated in line with the Turkish regulations. Please refer to Section 5.2 in VCS PD.	
SVS_{LT,y} - Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y	The SVS_{LT,y} parameter is monitored weekly. Therefore, the analysis of the related parameter is performed and recorded on a weekly basis for each animal farms in line with the Annex 2: 'Method for determination for Volatile Solids in animal waste' and the Annex 4: 'Guidance on sample extraction and statistical procedures' of the approved baseline and monitoring CDM methodology of AM0073. Regarding the laboratory analysis of SVS_{LT,y} parameter, there are 3 monitoring equipment named muffle furnace, dry air sterilizer, and scale at the laboratory installed at the biogas plant. Laboratory personnel record the detailed analysis data to the laboratory logbook then transfer the data to electronic format. Recorded data is reported to the site engineer on a	Laboratory personnel / Site engineer

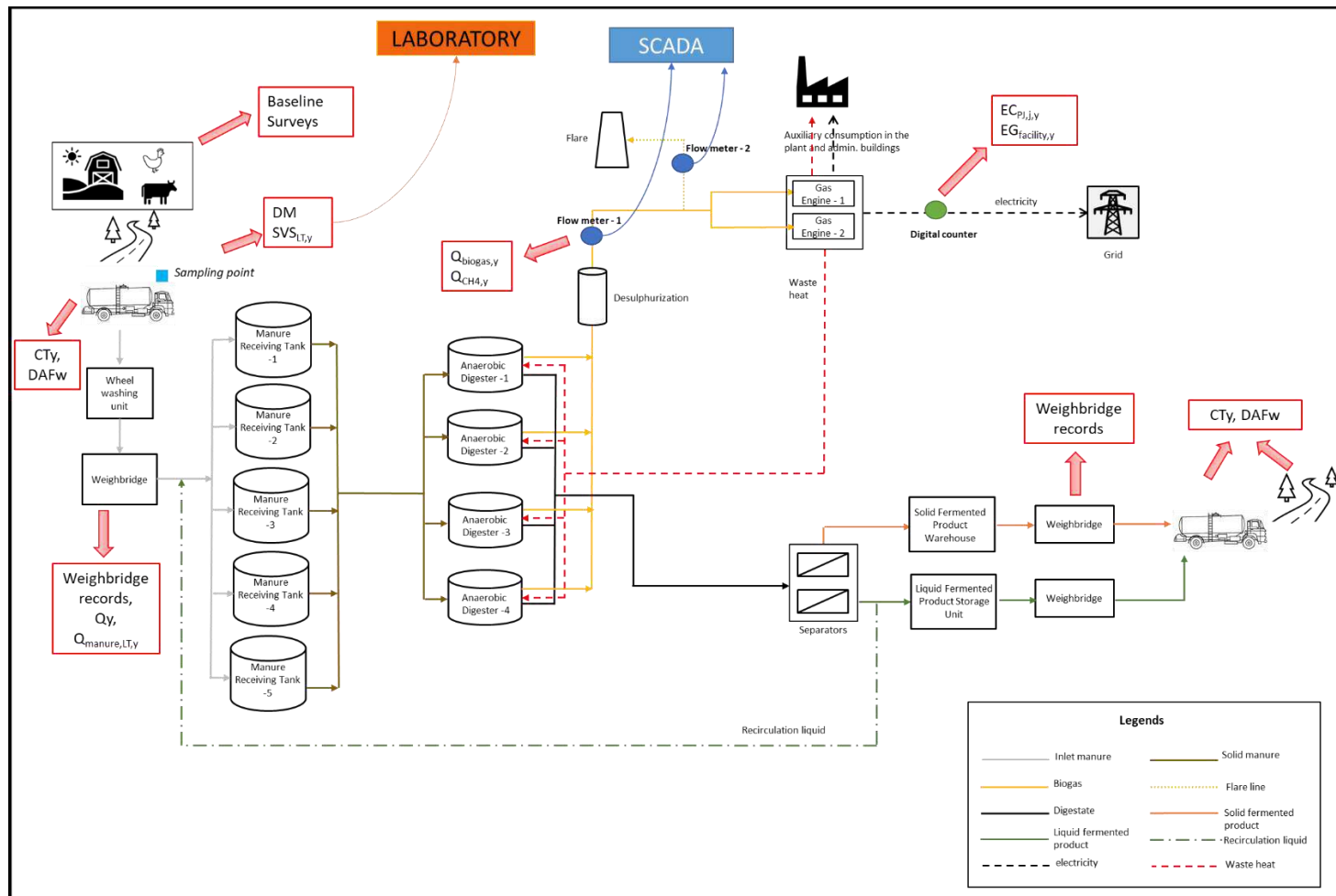
	weekly basis. The relevant monitoring equipment are calibrated by Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change. Please refer to Section 5.2 in VCS PD.	
$Q_{y,j}$ - Quantity of manure (wet basis) treated from livestock type LT at animal manure management system j	The quantity of manure on wet basis and other relevant details are recorded by the weighbridge operator and reported to the site engineer. Daily recordings are aggregated monthly and recorded. The weighbridge is calibrated in line with the Turkish regulations. Please refer to Section 5.2 in VCS PD.	Weighbridge operator / Site engineer
$Q_{manure,LT,y}$ - Quantity of manure (dry basis) treated from livestock type LT at animal manure management system j	Quantity of manure on dry basis is monitored through the weighbridge installed at the entrance of the biogas plant and calculated by taking into the dry matter (DM%) of the manure. The weighbridge is calibrated in line with the Turkish regulations. Please refer to Section 5.2 in VCS PD.	Weighbridge operator / Site engineer
DM - Residue remaining after water is removed from waste material by evaporation; dry matter	Regarding the laboratory analysis of DM parameter, there is one monitoring equipment named moisture analyzer at the laboratory installed at the biogas plant. Laboratory personnel record the detailed analysis data to the laboratory logbook then transfer the data to electronic format. Recorded data is reported to the site engineer on a	Laboratory personnel

	weekly basis. The relevant monitoring equipment is calibrated by Guidance prepared by Republic of Turkey Ministry of Environment, Urbanization and Climate Change. Please refer to Section 5.2 in VCS PD.	
DAF _w - Average incremental distance for raw solid manure transportation	Monitored through direct measurement of the distance between the collection point of the manure-to-manure discharge point.	Site manager

Above table lists all the parameters monitored on site. The Carbon Manager, as a PP also monitors;

- MCF_j parameter taken from the IPCC guidelines, as updated in July 2023;
- average technical transmission and distribution losses for providing electricity (TDL_{j,y}) sourced from the Turkish Electricity Transmission Corporation (TEİAŞ);
- and carbon emissions factor for the grid (EF_{grid,y}) sourced from the Turkish Ministry of Energy and Natural Resources.

SDG related parameters / indicators are also monitored by the PP. Amount of electricity supplied to the national grid (related to SDG 7), is monitored through the electricity bills and two-way electricity meters as indicated above for the EG_{facility,y} parameter as well. The amount of waste treated at the plant during the monitoring period (related to SDG 12) is monitored through the weighbridge recordings. The quantity of removed/avoided net GHG emission (related to SDG 13) is monitored through the ER calculations completed by the PP. Number of permanent and skilled job created (related to SDG 8) is monitored through the personnel affairs' official records. Project's participation in public-private partnerships for infrastructure is monitored through YEKDEM scheme records. All the recordings related to SDG contributions will also be archived electronically at least for five years after the end of the last crediting period.



Quality Control / Quality Assurance Procedures

Quality control and quality assurance procedures will guarantee the quality of monitored data.

Storing Process Variables and Recording Operational Data

All-important process variables and measurement values are recorded and stored by the Scada operator, site engineer, and site manager. The process variables can be displayed as needed. Data is monitored in line with the monitoring frequency of each variable/parameter. Monitored data is archived electronically and backed up every 24 hours online. Further, all data collected as part of monitoring will be archived electronically and be kept at least for five years after the end of the last crediting period. Written data and documents such as electricity invoices and diesel oil receipts are copied and archived and will be kept at least for five years after the end of the last crediting period.

Staff Trainings

As competency of staff involved in the monitoring activities is highly important, necessary trainings are provided by the project owner. Each member of the monitoring staff is informed about technical/methodological requirements for conducting a reliable monitoring operation. Only the qualified staff members are given the responsibility of monitoring.

Procedure in case of failure

If a monitoring equipment is in fault, it will be immediately repaired or replaced with another calibrated equipment by a professional staff. In the meantime, for the continuously monitored parameters, monitoring should be resumed via a backup equipment if possible.

Sampling Plan

Applied methodology requires testing according to the guideline in annex 2 of AM0073. Consequently, sampling procedure is conducted in line with the Annex 4 of the methodology AM0073. According to this guidance, the liquid material is collected using continuous-flow samples at the entrance or exit point of the pertinent treatment stage; samples are collected in clean wide-mouth glass bottles; are analyzed as soon as possible; and storage of the samples is performed at 4°C. Laboratory staff ensures that the suspended matter does not adhere to the walls, prior to the analysis procedure and the dry matter content is determined after oven drying at 103 °C for 24 hours or until constant weight is obtained. Such procedures are applied to the parameters subject to essays described in annex 2 of the methodology AM0073. The laboratory staff is responsible to adhere these guides and the site manager is responsible for ensuring the laboratory staff is trained to properly carry out the sampling procedure.

Emergency Preparedness

Site Manager ensures that the project personnel acquire the skills to carry out emergency tasks properly through capacity building practices and trainings.

The whole staff receives health and safety trainings to minimize exposure to workplace hazards. At every shift, at least one trained technical personnel are present. All relevant information such as data files, maintenance, defect records and other monitoring related records are kept at a designated location in a transparent and organized manner.

APPENDIX-1⁷⁹

Table 7: Name of Farms Supplying Manure to the Project, Depths of their Lagoons and Retention Periods

Farm Name	Lagoon Depth (in meters)	Retention period (in months)
Ali Bey Çiftliği	5	3
Erol Yılmaz	3,6	1
Bey Besi (Veysel Karabey)	4	1.5-2
Şenol Yılmaz Besi Çiftliği	3,6	1.5
Saygın Bey (Veysel Karabey)	8	1-2
Aytekin Besi	3,5	1
has besi	6	1.5-2
izzet alp	4	1-2
yılmazlar besi (kerim yılmaz)	5	1.5-2

⁷⁹ These are the data available in time of validation may be subjected to change in time of verification.

APPENDIX-2



Document 1: Energy Generation License of the Elmadağ Biogas Power Plant from the Republic of Turkey Energy Market Regulatory Authority (EPDK)

APPENDIX-3

**T.C.
ENERJİ VE TABİİ KAYNAKLAR
BAKANLIĞI**

KISMİ KABUL TUTANAĞI

**ABRAK ENERJİ ELEKTRİK ÜRETİM A. Ş.
ELMADAĞ BES**

**1 x (1,540 MW_m/ 1,203 MW_e)
GM1 No'lu
Gaz Motoru-Generatör Grubu**

LİSANS TARİHİ VE NUMARASI	: 01.08.2019 – EÜ/8760-1/04286
LİSANS TAKİ TESİS İLİ	: ANKARA
LİSANS TAKİ TESİS TİPİ	: YENİLENEBİLİR ENERJİ
LİSANS TOPLAM KURULU GÜCÜ	: 3,080 MW _m /2,406 MW _e
LİSANS ÜNİTE SAYISI VE KURULU GÜCÜ	: 2x(1,540 MW _m /1,203 MW _e)
KABULE ESAS TOPLAM KURULU GÜCÜ	: 1,540 MW _m /1,203 MW _e
KABULE ESAS ÜNİTE SAYISI VE KURULU GÜCÜ	: 1x(1,540 MW _m /1,203 MW _e)
BU KABUL İLE ULAŞILAN İŞLETME GÜCÜ	: 1,540 MW _m /1,203 MW _e
PROJE ONAY TARİH VE SAYISI	: 01.12.2020 / 38159

31.12.2020

**Document 2: Certificate of Provisional Acceptance to Generate Electricity of the Elmadag
Biogas Power Plant**

APPENDIX-4



Document 3: Environmental Impact Assessment of the Elmadag Biogas Power Plant